

L1 Logistics and Introduction

Dr. Zixin Zhong

Data Science and Analytics Thrust
Information Hub
Hong Kong University of Science and Technology (Guangzhou)

January 16, 2026

Grading scheme

- **Midterm 20 Points:** 28 March (Sat of Week # 7)
- **Written assignments 40 Points:** group project
- **Final exam 40 Points:** [TBD]
- **Bonus 5 points:**
 - 3 points: proposing good questions/answers, spotting nontrivial mistakes in slides, etc.
 - 2 points: 1 point for 5 lab attendance.

Letter Grade	Raw Mark Range (Reference)
A	[85, 100]
B	[70, 85)
C	[55, 70)
D	[40, 55)
F	< 40



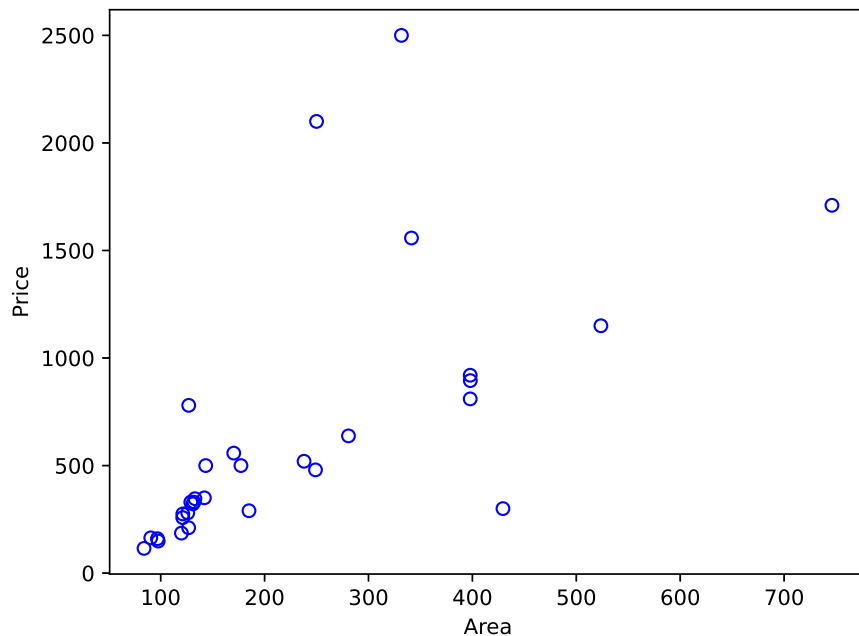
Supervised Learning



Task: housing price prediction

- Given a dataset containing n samples

$$\left(x^{(1)}, y^{(1)}\right), \left(x^{(2)}, y^{(2)}\right), \left(x^{(3)}, y^{(3)}\right), \dots, \left(x^{(n)}, y^{(n)}\right)$$

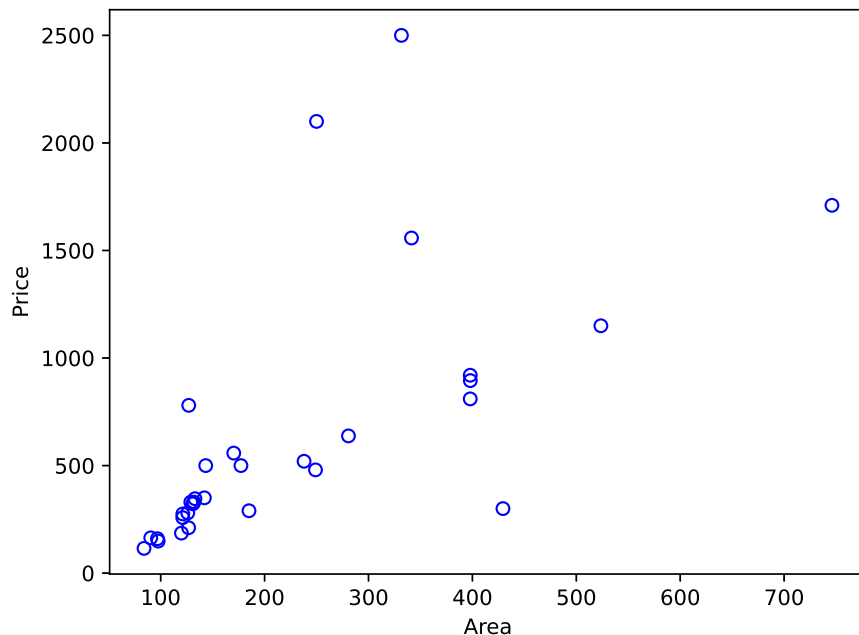


Task: housing price prediction

- Given a dataset containing n samples

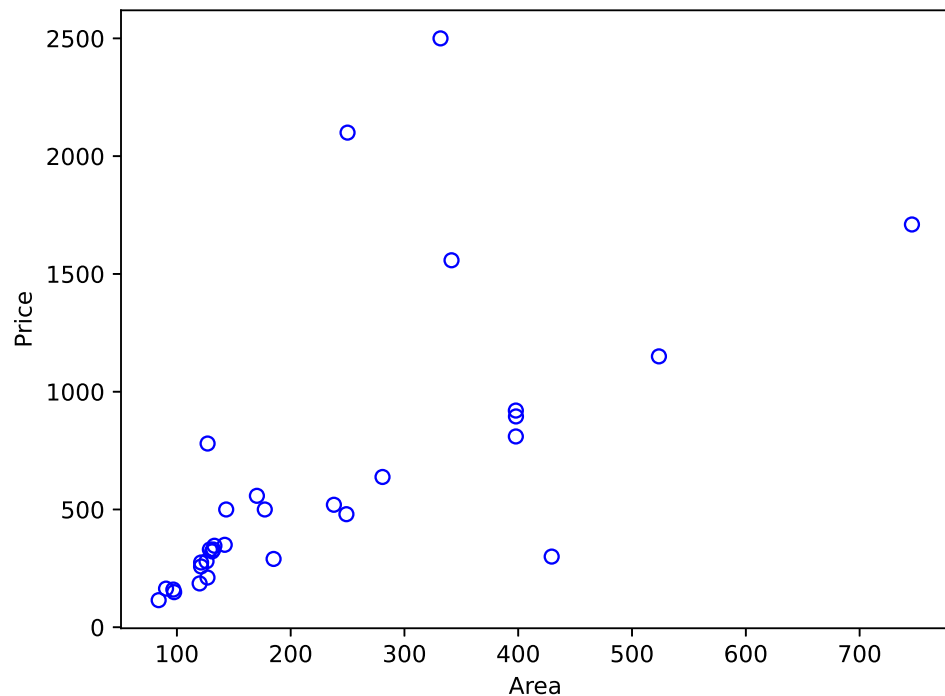
$$\left(x^{(1)}, y^{(1)}\right), \left(x^{(2)}, y^{(2)}\right), \left(x^{(3)}, y^{(3)}\right), \dots, \left(x^{(n)}, y^{(n)}\right)$$

- Task:** if a residence has $x = 320$ square kilometers, can we predict its price?



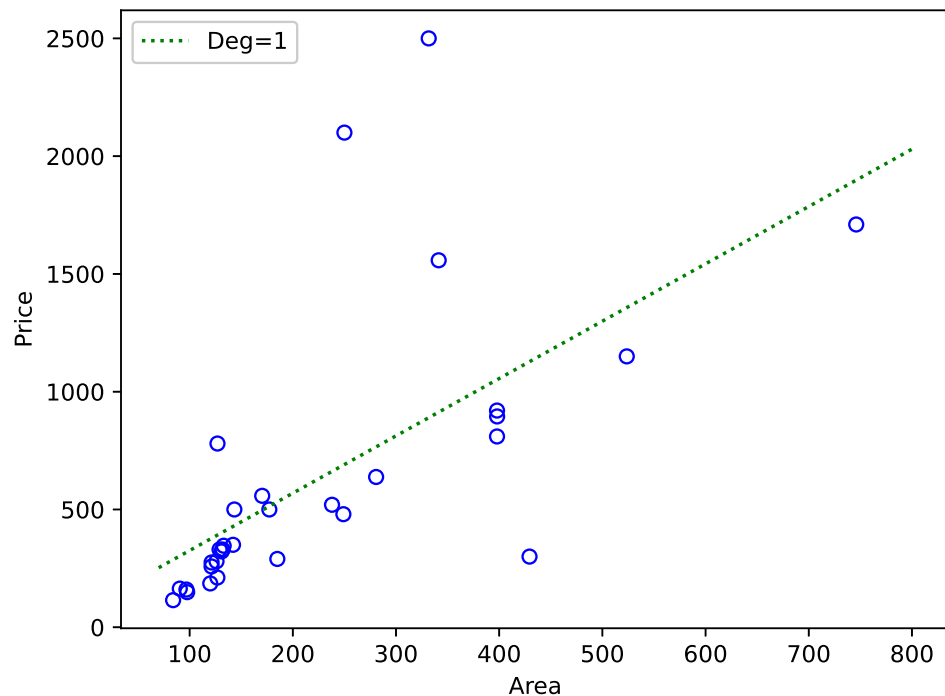
Task: housing price prediction

- Linear regression (polynomial regression with degree 1):
Assume $y = p_1(x) = ax + b$ and guess the values of a and b .



Task: housing price prediction

- Linear regression (polynomial regression with degree 1):
Assume $y = p_1(x) = ax + b$ and guess the values of a and b .



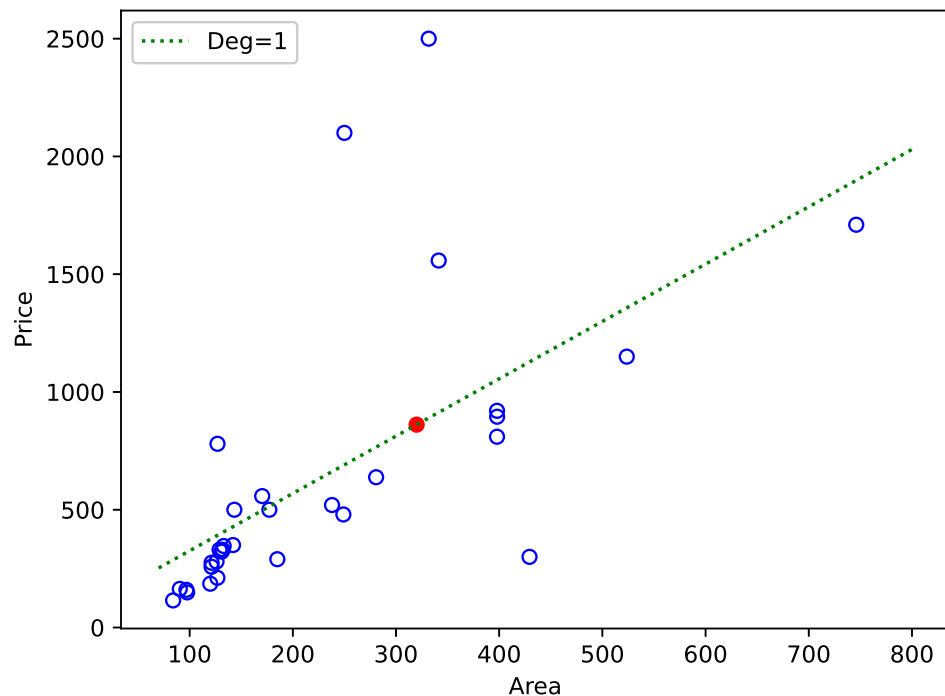
$$a = 2.43, b = 82.36$$

$$p_1(x) = 2.43x + 82.36$$



Task: housing price prediction

- Linear regression (polynomial regression with degree 1):
Assume $y = p_1(x) = ax + b$ and guess the values of a and b .



$$a = 2.43, b = 82.36$$

$$p_1(x) = 2.43x + 82.36$$

$$p_1(320) = 861.29$$

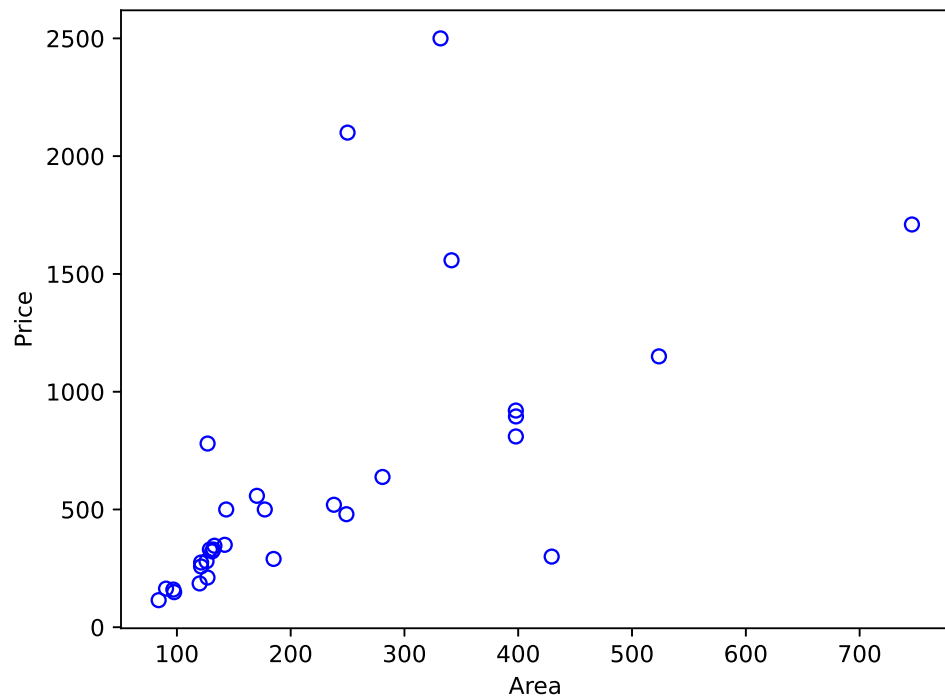
- If a residence has $x = 320$ square kilometers, we predict its price as $y = 861.29$ CNY.



Task: housing price prediction

- Polynomial regression with degree 2:

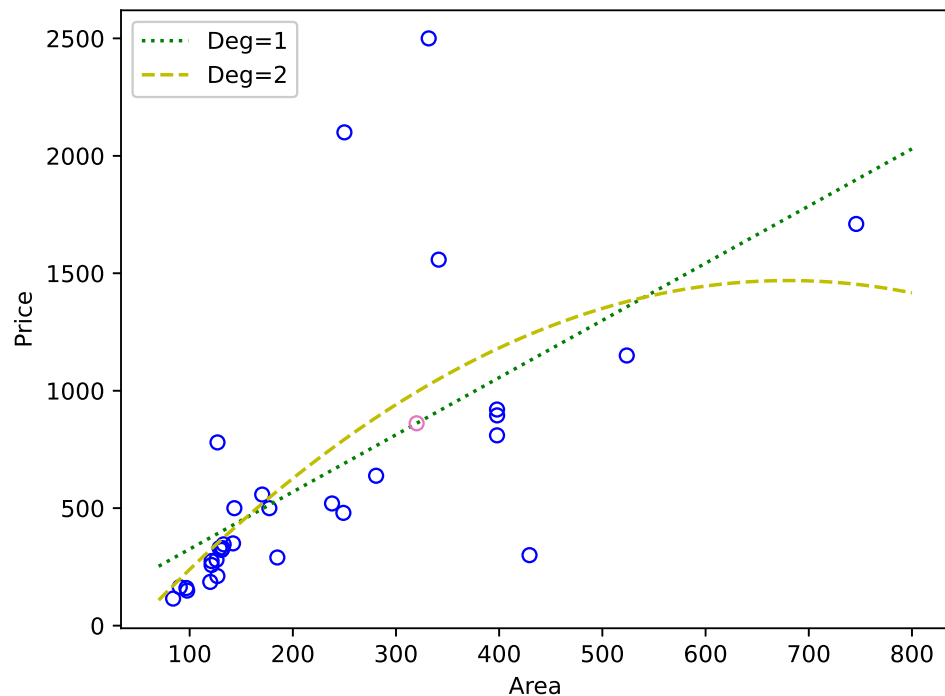
Assume $y = p_2(x) = ax^2 + bx + c$ and guess the values of a , b and c .



Task: housing price prediction

- Polynomial regression with degree 2:

Assume $y = p_2(x) = ax^2 + bx + c$ and guess the values of a , b and c .



$$a = -0.0037, b = 4.97, c = -221.8$$

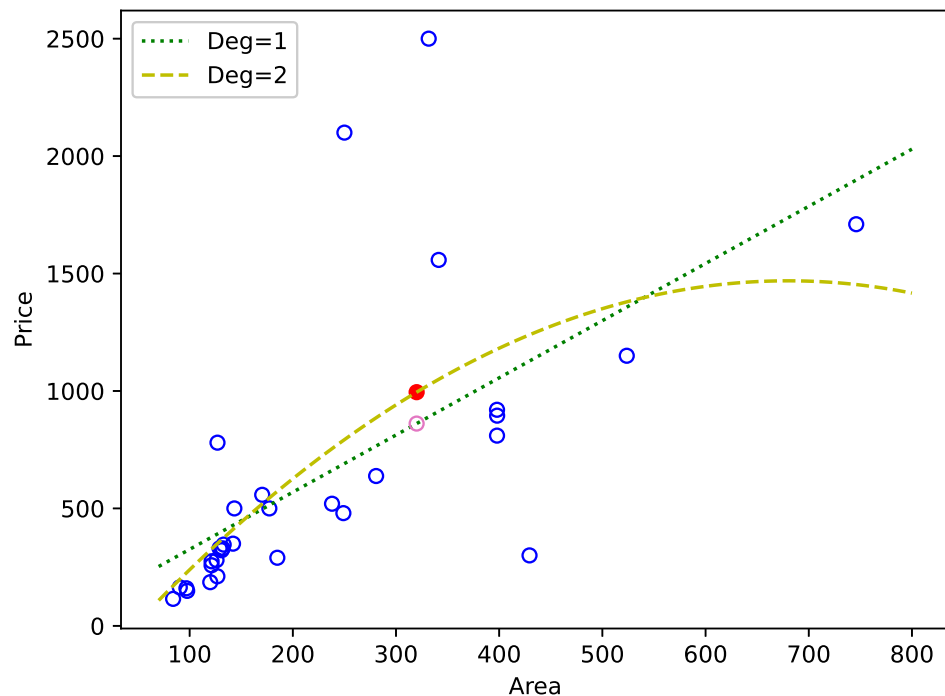
$$p_2(x) = -0.0037x^2 + 4.97x - 221.8$$



Task: housing price prediction

- Polynomial regression with degree 2:

Assume $y = p_2(x) = ax^2 + bx + c$ and guess the values of a , b and c .



$$a = -0.0037, b = 4.97, c = -221.8$$

$$p_2(x) = -0.0037x^2 + 4.97x - 221.8$$

$$p_2(320) = 994.94$$

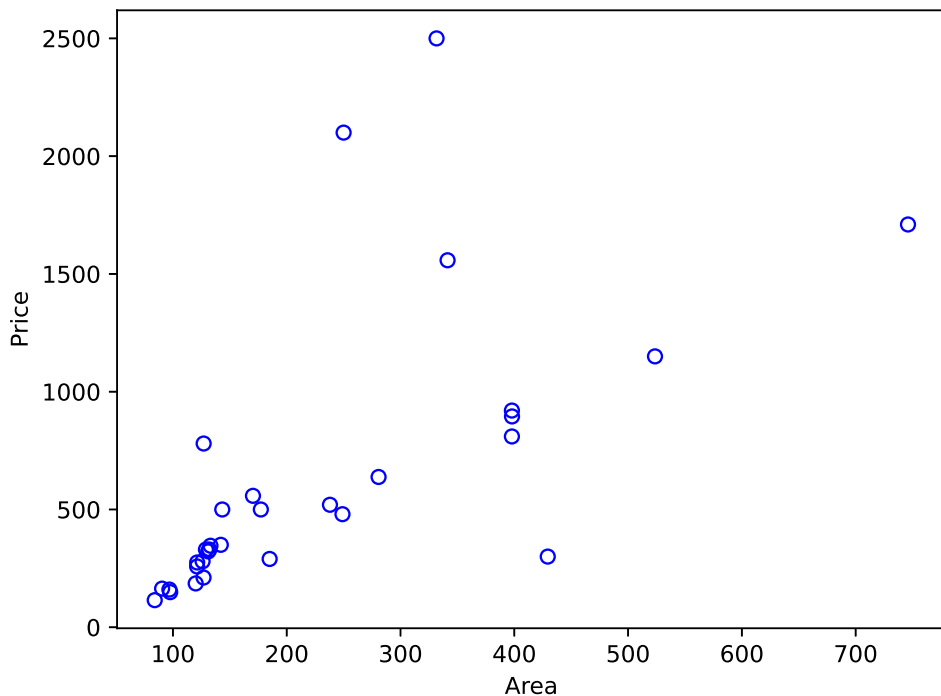
- If a residence has $x = 320$ square kilometers, we predict its price as $y = 994.94$ CNY.



Task: housing price prediction

- Polynomial regression with degree 3:

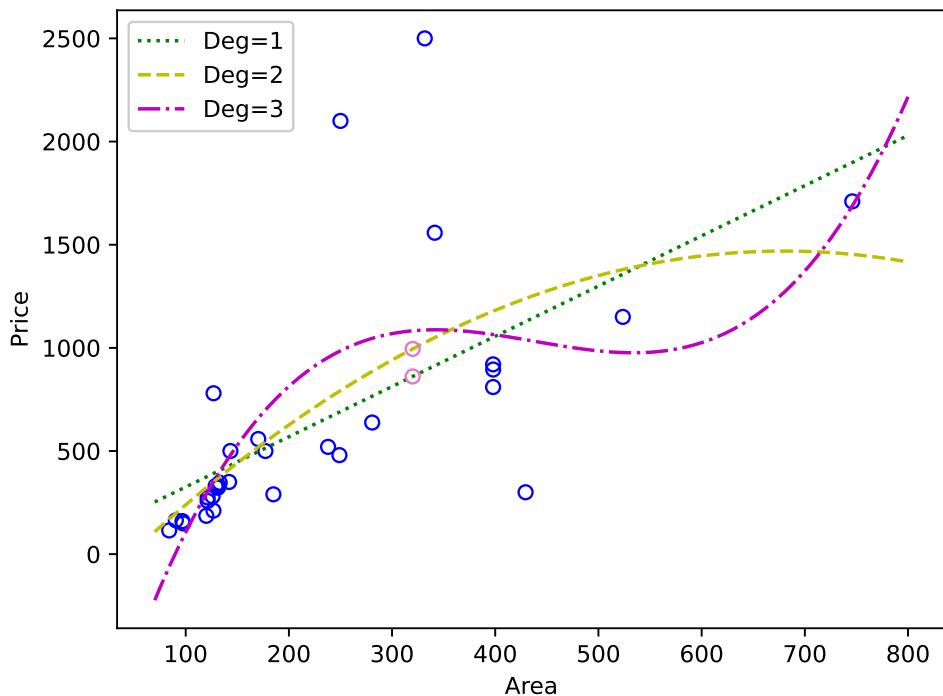
Assume $y = p_3(x) = ax^3 + bx^2 + cx + d$ and guess the values of a , b , c and d .



Task: housing price prediction

- Polynomial regression with degree 3:

Assume $y = p_3(x) = ax^3 + bx^2 + cx + d$ and guess the values of a , b , c and d .



$$a = 3.16 * 10^{-5}, b = -0.042,$$

$$c = 17.33, d = -1243$$

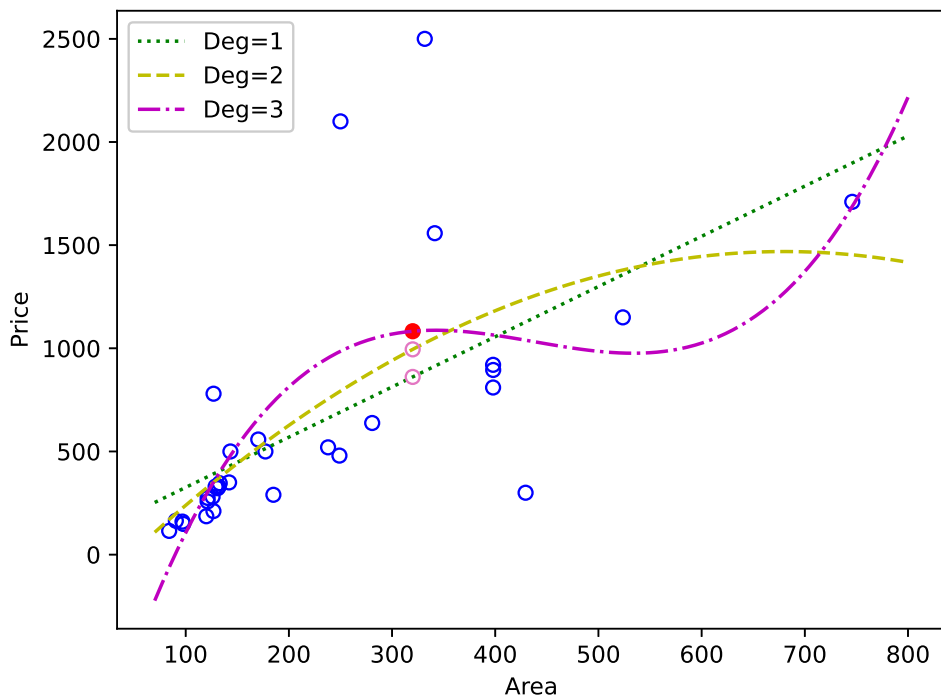
$$p_3(x) = 3.16 * 10^{-5}x - 0.042x + 17.33x - 1243$$



Task: housing price prediction

- Polynomial regression with degree 3:

Assume $y = p_3(x) = ax^3 + bx^2 + cx + d$ and guess the values of a , b , c and d .



$$a = 3.16 * 10^{-5}, b = -0.042,$$

$$c = 17.33, d = -1243$$

$$p_3(x) = 3.16 * 10^{-5}x - 0.042x + 17.33x - 1243$$

$$p_3(320) = 1082.80$$

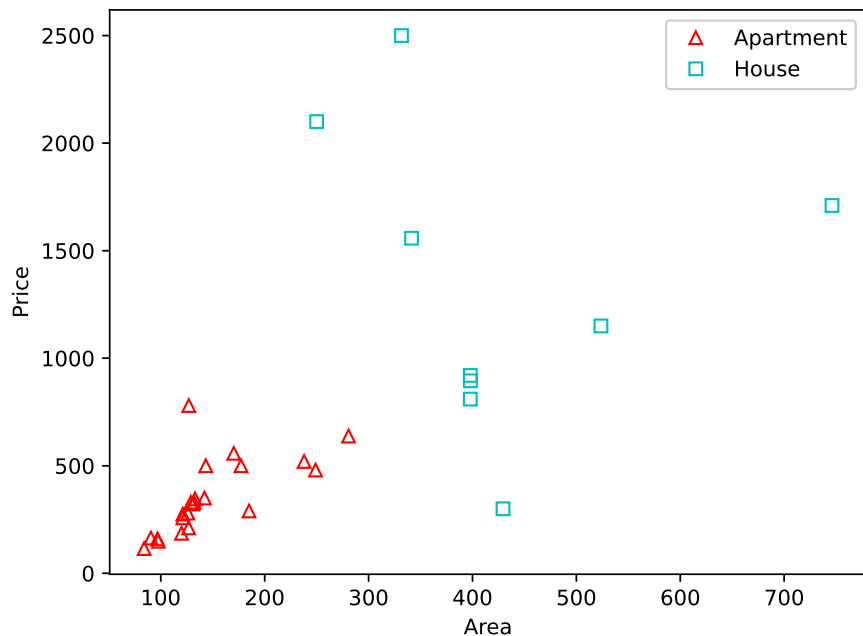
- If a residence has $x = 320$ square kilometers, we predict its price as $y = 1082.80$ CNY.



Task: housing type prediction

- Given a dataset containing n samples ($z \in \{0, 1\}$)

$$\left(x^{(1)}, y^{(1)}, z^{(1)}\right), \left(x^{(2)}, y^{(2)}, z^{(2)}\right), \left(x^{(3)}, y^{(3)}, z^{(3)}\right), \dots, \left(x^{(n)}, y^{(n)}, z^{(n)}\right)$$

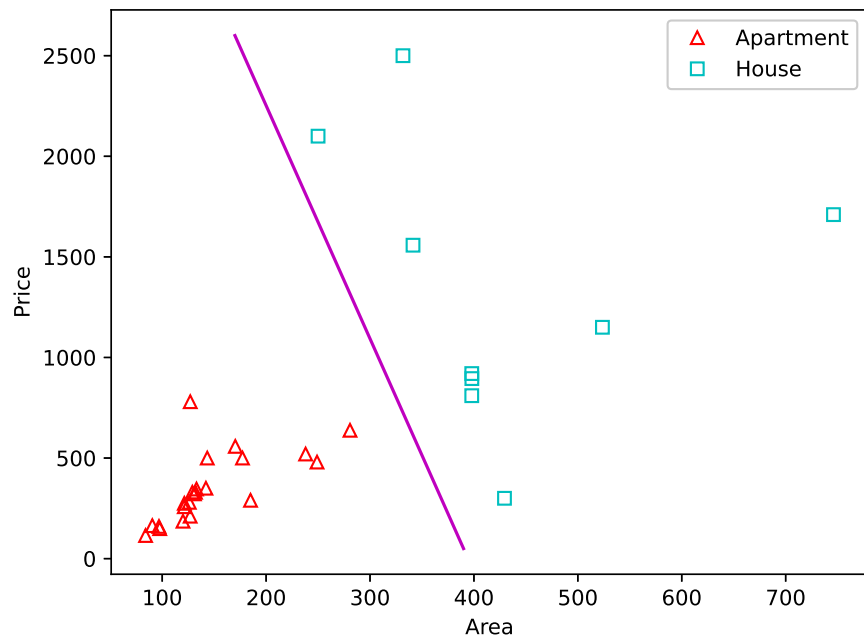


Task: housing type prediction

- Given a dataset containing n samples ($z \in \{0, 1\}$)

$$\left(x^{(1)}, y^{(1)}, z^{(1)}\right), \left(x^{(2)}, y^{(2)}, z^{(2)}\right), \left(x^{(3)}, y^{(3)}, z^{(3)}\right), \dots, \left(x^{(n)}, y^{(n)}, z^{(n)}\right)$$

- Task:** if a residence has $x = 320$ square kilometers and worth $y = 1080.20$ CNY, can we predict its type z ?

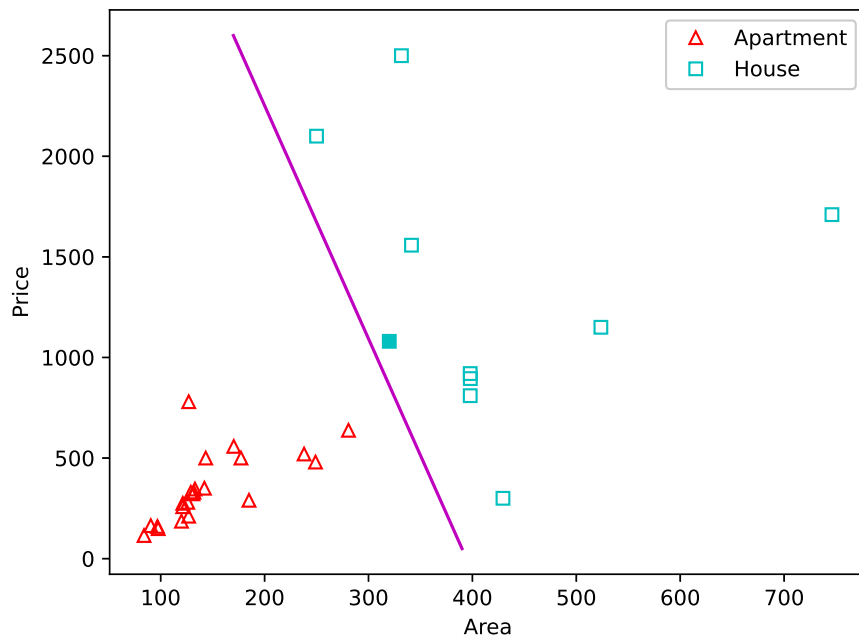


Task: housing type prediction

- Given a dataset containing n samples ($z \in \{0, 1\}$)

$$\left(x^{(1)}, y^{(1)}, z^{(1)}\right), \left(x^{(2)}, y^{(2)}, z^{(2)}\right), \left(x^{(3)}, y^{(3)}, z^{(3)}\right), \dots, \left(x^{(n)}, y^{(n)}, z^{(n)}\right)$$

- Task:** if a residence has $x = 320$ square kilometers and worth $y = 1080.20$ CNY, can we predict its type z ?



Task comparison: regression vs classification

- **Regression:** the prediction result is a **continuous** variable
 - ▶ e.g., price prediction
 $(x = \text{Area}) \rightarrow (y = \text{Price})?$
- **Classification:** the prediction result is a **discrete** variable
 - ▶ e.g., type prediction
 $(x = \text{Area}, y = \text{Price}) \rightarrow (z = \text{Type})?$



Unsupervised Learning

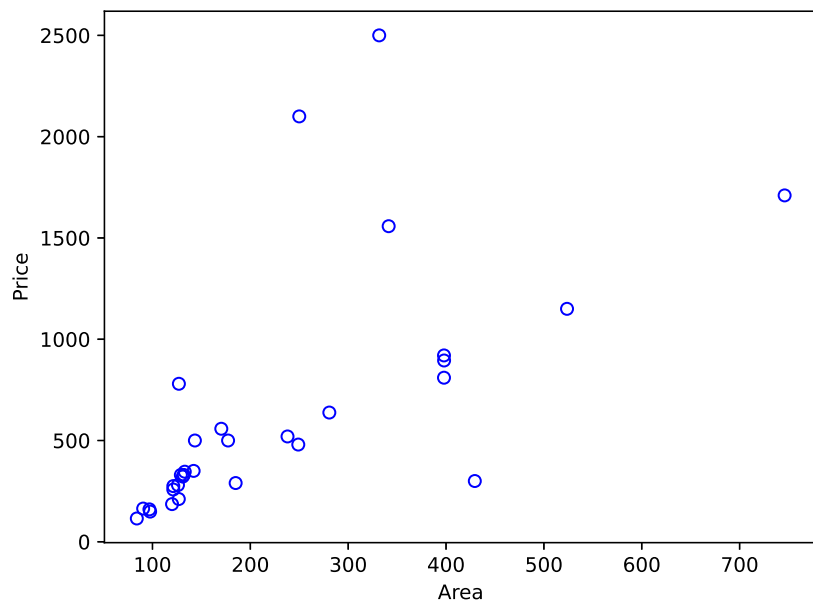


Task: housing type prediction

- Given a dataset containing n samples

$$\left(x^{(1)}, y^{(1)}\right), \left(x^{(2)}, y^{(2)}\right), \left(x^{(3)}, y^{(3)}\right), \dots, \left(x^{(n)}, y^{(n)}\right)$$

- Task:** can we learn the type of each residence?

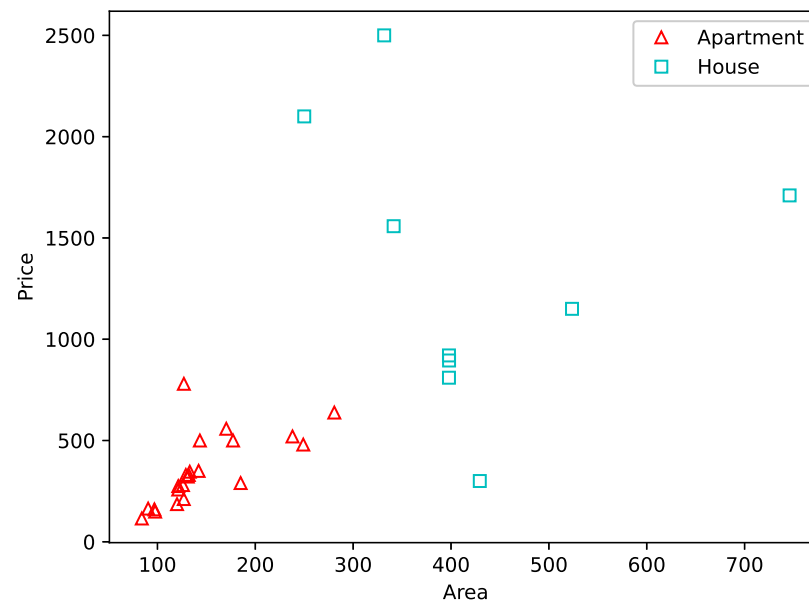
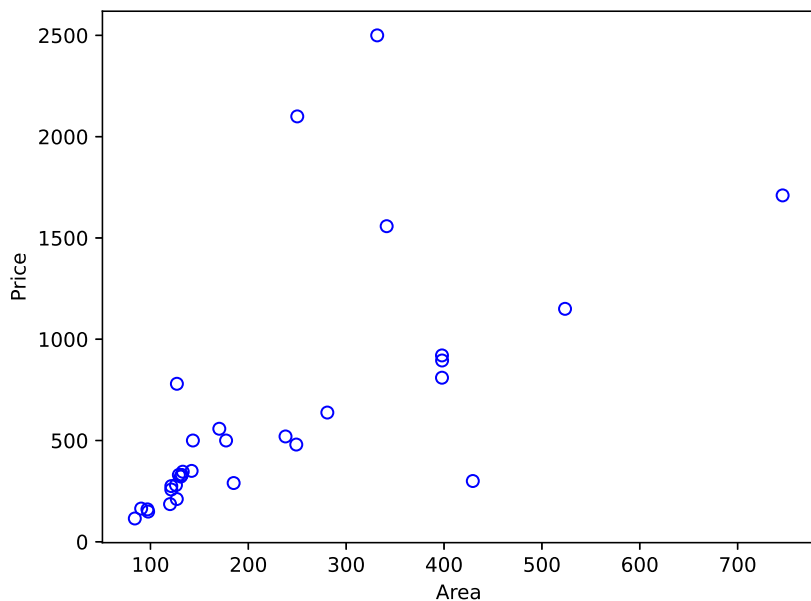


Task: housing type prediction

- Given a dataset containing n samples

$$\left(x^{(1)}, y^{(1)}\right), \left(x^{(2)}, y^{(2)}\right), \left(x^{(3)}, y^{(3)}\right), \dots, \left(x^{(n)}, y^{(n)}\right)$$

- Task:** can we learn the type of each residence?

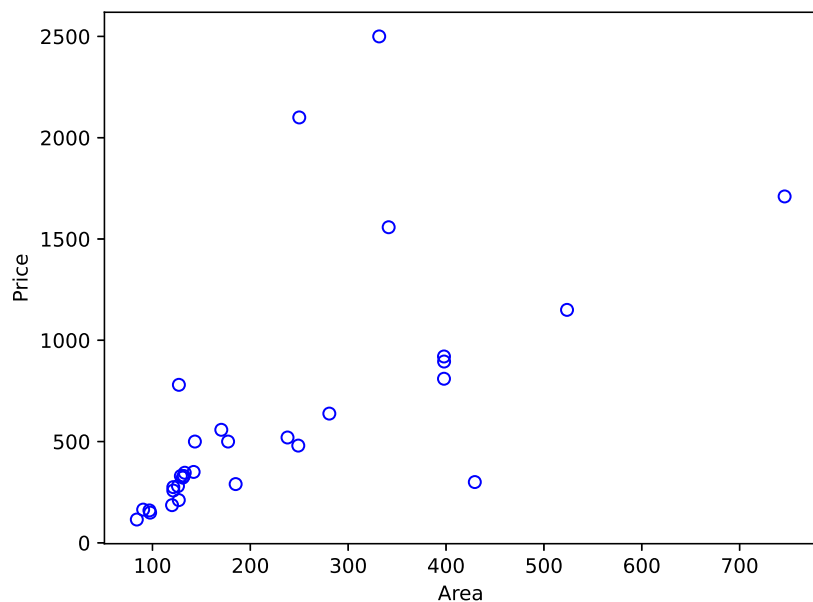


Clustering

- Given a dataset containing n samples

$$\left(x^{(1)}, y^{(1)}\right), \left(x^{(2)}, y^{(2)}\right), \left(x^{(3)}, y^{(3)}\right), \dots, \left(x^{(n)}, y^{(n)}\right)$$

- Task (vague):** find interesting structures in the data

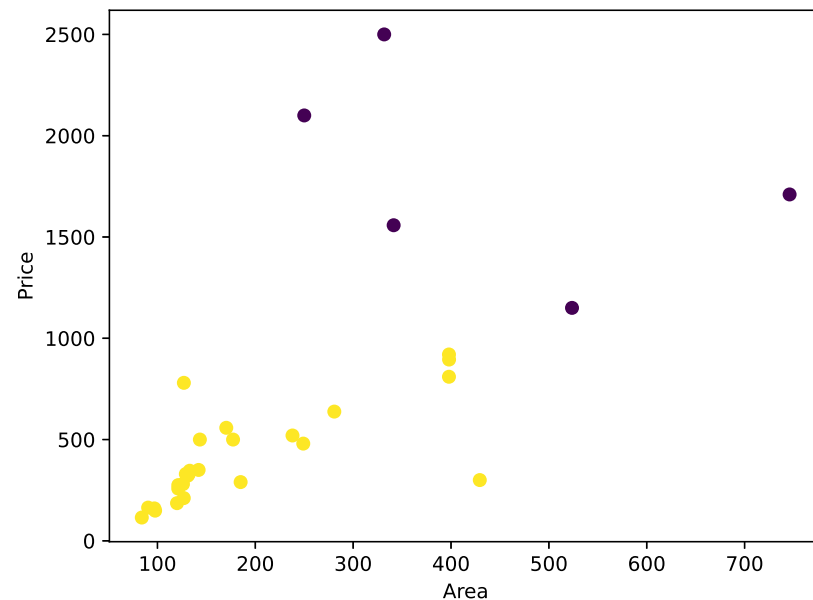
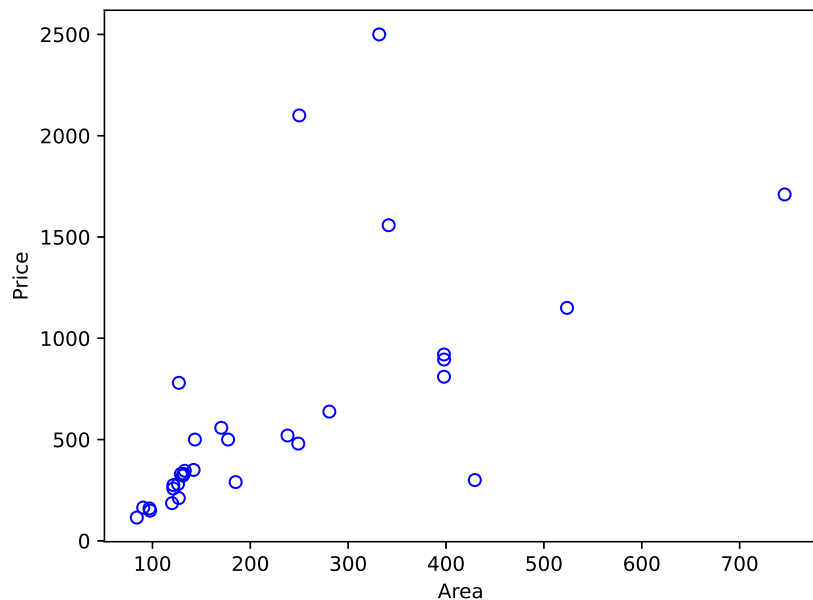


Clustering

- Given a dataset containing n samples

$$\left(x^{(1)}, y^{(1)}\right), \left(x^{(2)}, y^{(2)}\right), \left(x^{(3)}, y^{(3)}\right), \dots, \left(x^{(n)}, y^{(n)}\right)$$

- Task (vague):** find interesting structures in the data

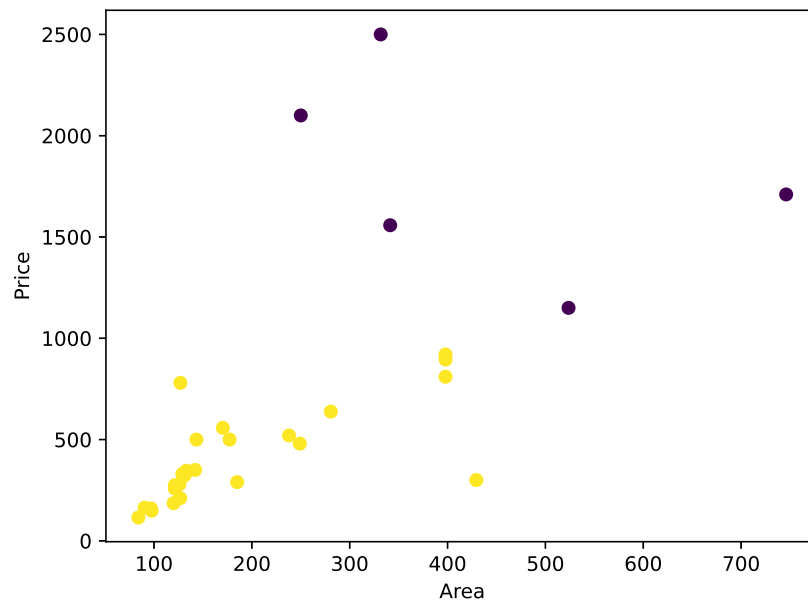
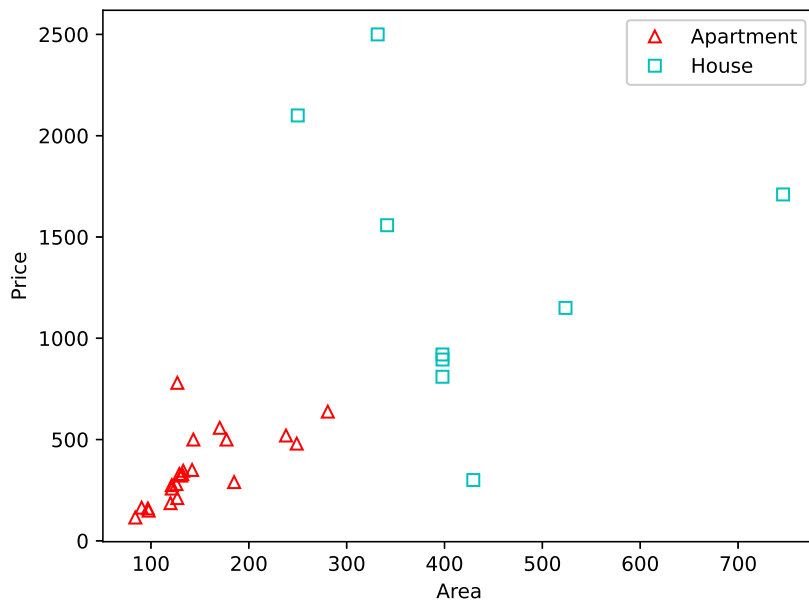


Clustering

- Given a dataset containing n samples

$$\left(x^{(1)}, y^{(1)}\right), \left(x^{(2)}, y^{(2)}\right), \left(x^{(3)}, y^{(3)}\right), \dots, \left(x^{(n)}, y^{(n)}\right)$$

- Task (vague):** find interesting structures in the data

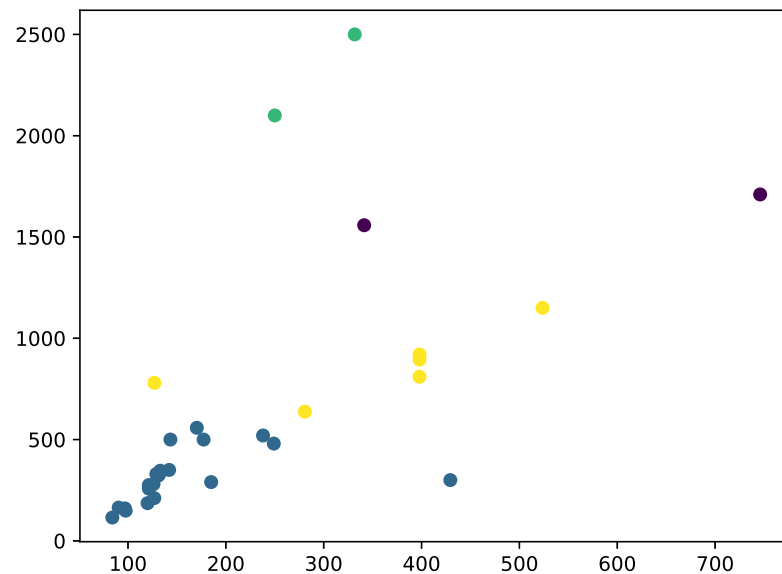
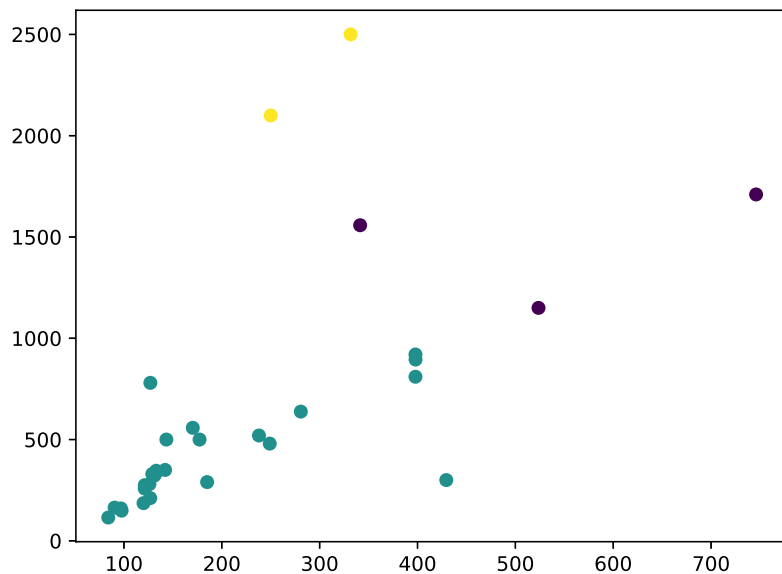


Clustering

- Given a dataset containing n samples

$$\left(x^{(1)}, y^{(1)}\right), \left(x^{(2)}, y^{(2)}\right), \left(x^{(3)}, y^{(3)}\right), \dots, \left(x^{(n)}, y^{(n)}\right)$$

- Task (vague):** find interesting structures in the data



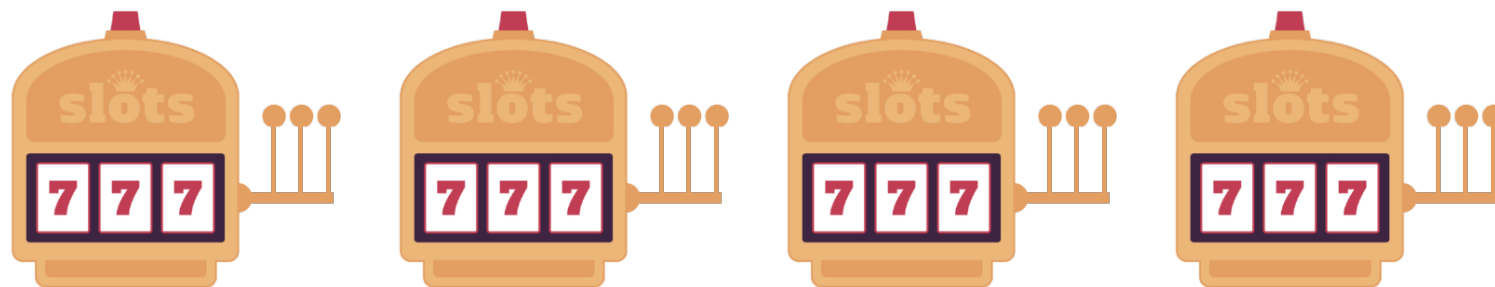
Reinforcement Learning



Task: win more

- We pull one of the 4 arms of a slot machine at each round t

$L = 4$

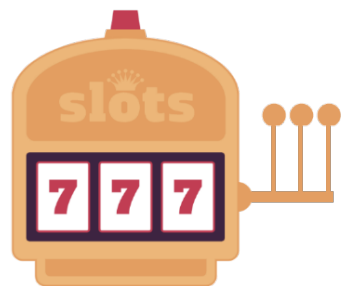


Task: win more

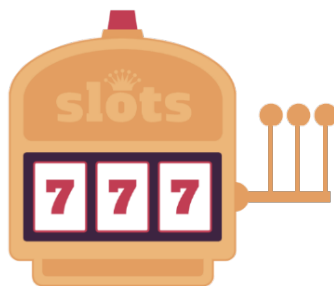
- We pull one of the 4 arms of a slot machine at each round t
- Each arm i yields a random reward $W_t(i)$ at round t

$L = 4$

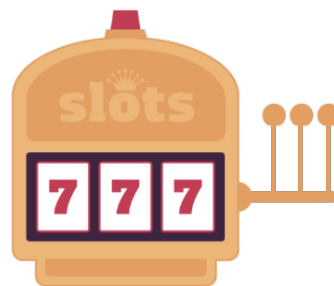
Reward



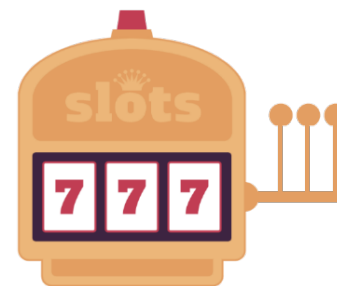
$W_t(1)$



$W_t(2)$



$W_t(3)$

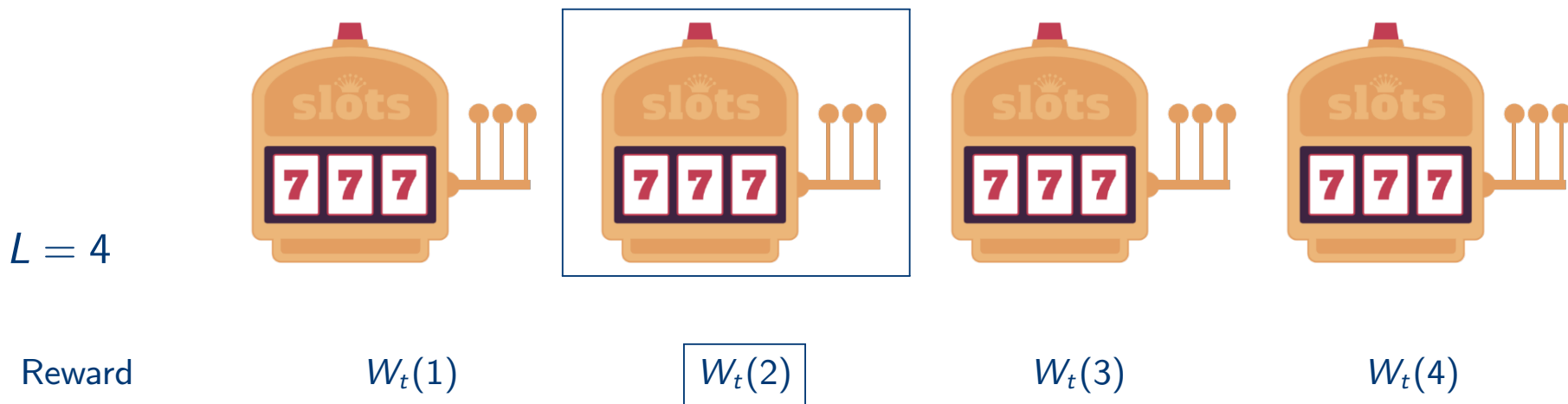


$W_t(4)$



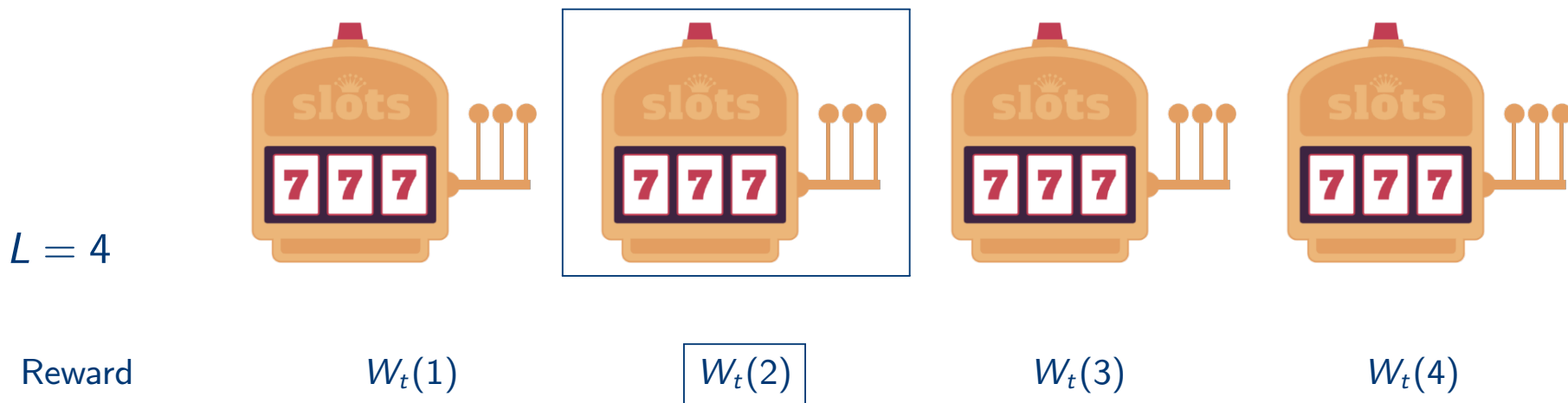
Task: win more

- We pull one of the 4 arms of a slot machine at each round t
- Each arm i yields a **random reward** $W_t(i)$ at round t
- At time t , if we pull arm 2, we earn $W_t(2)$

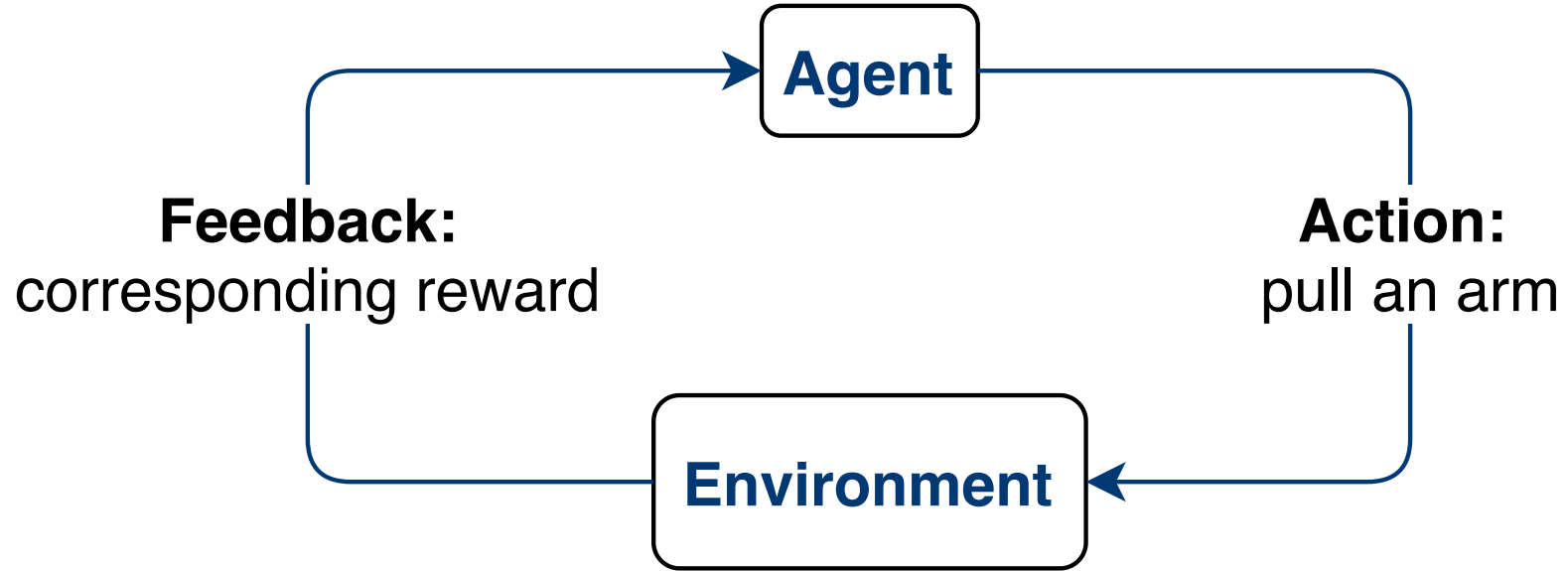


Task: win more

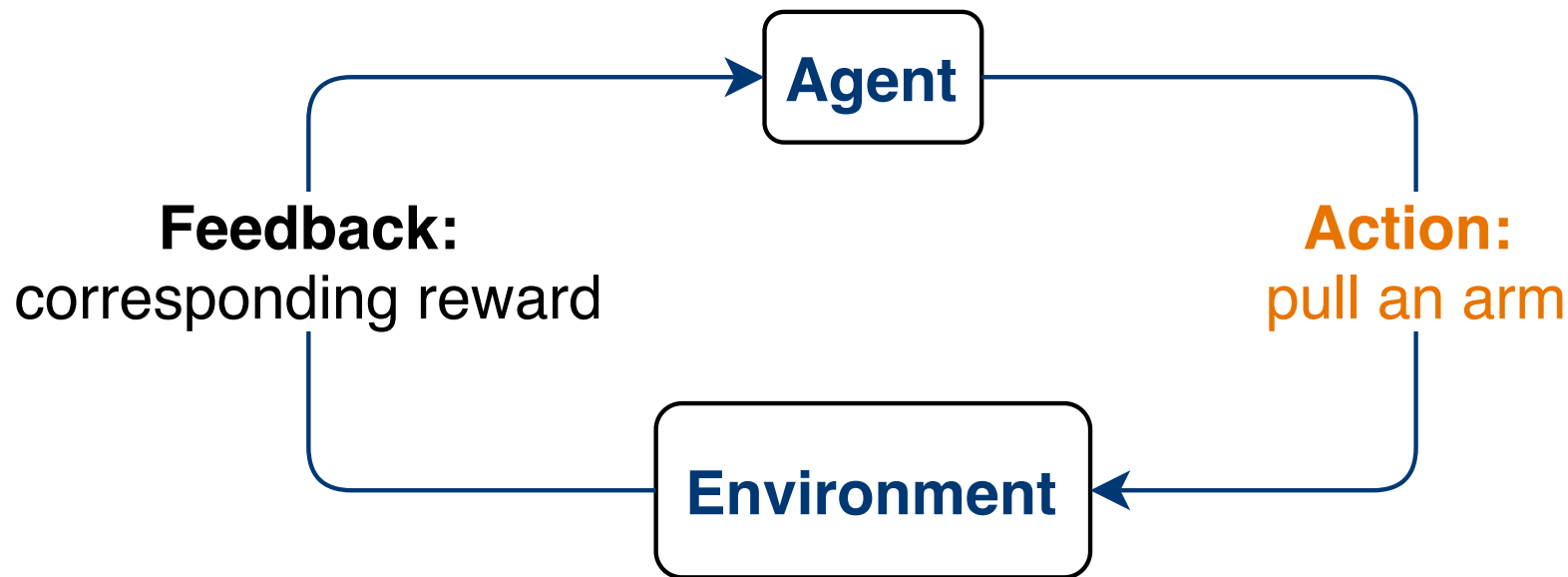
- We pull one of the 4 arms of a slot machine at each round t
- Each arm i yields a **random reward** $W_t(i)$ at round t
- At time t , if we pull arm 2, we earn $W_t(2)$
- **Task:** how can we earn as much as possible after 100 rounds?



Multi-armed bandit problem (MAB)



Multi-armed bandit problem (MAB)



- Instead of labeled/unlabeled data, a **reinforcement learning** algorithm learns from environment feedback.



Taxonomy of Machine Learning (A Simplistic View)

♠ What type of data?

- Supervised learning - **labeled data**: e.g., prediction
- Semi-supervised learning
- Unsupervised learning - **unlabeled data**: e.g., clustering
- Reinforcement learning - **environment feedback**: e.g., multi-armed bandit

♠ When do we collect data?

- Offline learning 先收集数据, 再训练
- Online learning 边来数据边训练



Outline

- 1 Logistics
- 2 Introduction to Machine Learning
 - Definition and taxonomy
- 3 Mathematical tools
 - Set and function
 - Probability and estimation
 - Systems of linear equations



Sets

- A set S is an **unordered collection of objects**.
- $S = \{1, 2, 3, 4, 5, 6\}$ is the possible outcomes of the toss of a die.
- $S = [a, b] = \{x \in \mathbb{R} : a \leq x \leq b\}$ is the set of all numbers from a to b inclusive.
- $S = (a, b] = \{x \in \mathbb{R} : a < x \leq b\}$ is the set of all numbers from a to b , excluding a including b .
- $\mathbb{R}$ is the set of all real numbers.
- $\mathbb{R}^d$ is the set of all real vectors of length d .

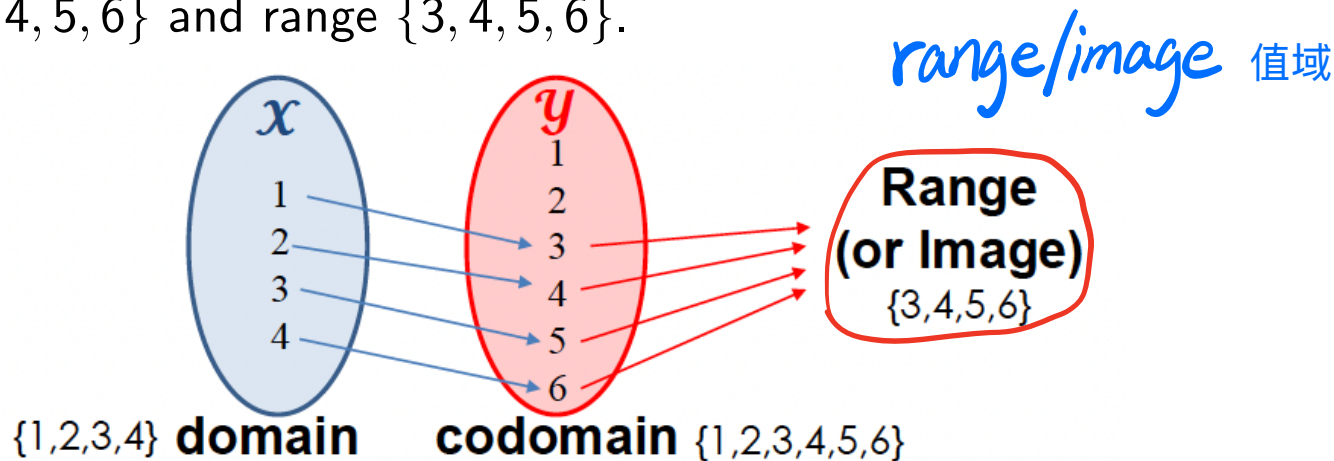


Functions

- A function f is a **map** from a set X to another set Y . We write this as

$$f : X \rightarrow Y.$$

- For example, the function $f : \mathbb{R} \rightarrow [0, \infty)$ could be given by the recipe $f(x) = x^2$.
- The set of inputs is called the **domain**; the set of possible outputs is called the **codomain**; the set $\{f(x) : x \in X\}$ is called the **range** (or **image**).
- For example $f : \{1, 2, 3, 4\} \rightarrow \{1, 2, 3, 4, 5, 6\}$ given by the recipe $f(x) = x + 2$ has codomain $\{1, 2, 3, 4, 5, 6\}$ and range $\{3, 4, 5, 6\}$.



Functions

Definition 3.1

Let $f(x)$ be a continuous function defined on domain $\mathbb{R}$.

- $f(x)$ is **differentiable** if $f'(x) = \lim_{x \rightarrow c} \frac{f(x) - f(c)}{x - c}$ exists for all $x \in \mathbb{R}$.
- $f(x)$ is **convex** if $f(\lambda a + (1 - \lambda)b) \leq \lambda f(a) + (1 - \lambda)f(b)$ for all $a, b \in \mathbb{R}$;
- $f(x)$ is **strictly convex** if $f(\lambda a + (1 - \lambda)b) < \lambda f(a) + (1 - \lambda)f(b)$ for all $a, b \in \mathbb{R}$;
- $f(x)$ is **concave** if $f(\lambda a + (1 - \lambda)b) \geq \lambda f(a) + (1 - \lambda)f(b)$ for all $a, b \in \mathbb{R}$;
- $f(x)$ is **strictly concave** if $f(\lambda a + (1 - \lambda)b) > \lambda f(a) + (1 - \lambda)f(b)$ for all $a, b \in \mathbb{R}$;



Functions

Definition 3.1

Let $f(x)$ be a continuous function defined on domain $\mathbb{R}$.

- $f(x)$ is **differentiable** if $f'(x) = \lim_{x \rightarrow c} \frac{f(x) - f(c)}{x - c}$ exists for all $x \in \mathbb{R}$.
- $f(x)$ is **convex** if $f(\lambda a + (1 - \lambda)b) \leq \lambda f(a) + (1 - \lambda)f(b)$ for all $a, b \in \mathbb{R}$;
- $f(x)$ is **strictly convex** if $f(\lambda a + (1 - \lambda)b) < \lambda f(a) + (1 - \lambda)f(b)$ for all $a, b \in \mathbb{R}$;
- $f(x)$ is **concave** if $f(\lambda a + (1 - \lambda)b) \geq \lambda f(a) + (1 - \lambda)f(b)$ for all $a, b \in \mathbb{R}$;
- $f(x)$ is **strictly concave** if $f(\lambda a + (1 - \lambda)b) > \lambda f(a) + (1 - \lambda)f(b)$ for all $a, b \in \mathbb{R}$;

- $f(x)$ is differentiable and convex: x_0 such that $f'(x_0) = 0$ minimize function f .
- $f(x)$ is differentiable and concave: x_0 such that $f'(x_0) = 0$ maximize function f .



Linear functions

A function $f : \mathbb{R}^d \rightarrow \mathbb{R}$ is **linear** if it satisfies

- **(Homogeneity)** For any vector $\mathbf{x} \in \mathbb{R}^d$ and scalar $a \in \mathbb{R}$,

$$f(a \mathbf{x}) = a f(\mathbf{x})$$

- **(Additivity)** For any two vectors $\mathbf{x}, \mathbf{y} \in \mathbb{R}^d$,

$$f(\mathbf{x} + \mathbf{y}) = f(\mathbf{x}) + f(\mathbf{y})$$

Note that a linear function $f : \mathbb{R}^d \rightarrow \mathbb{R}$ must pass through the origin, i.e., $f(\mathbf{0}) = 0$ where $\mathbf{0} \in \mathbb{R}^d$ is the zero vector in d dimensions. **Why?**



Affine functions

- An affine function f is a linear function plus possibly a constant.
- More precisely, a function $f : \mathbb{R}^d \rightarrow \mathbb{R}$ is affine if it can be expressed as

$$f(\mathbf{x}) = \mathbf{a}^\top \mathbf{x} + b$$

↳ offset

for some vector $\mathbf{a} \in \mathbb{R}^d$ and some scalar $b \in \mathbb{R}$.

- The scalar b is called the bias or offset.

Example: The following function $f : \mathbb{R}^2 \rightarrow \mathbb{R}$ is affine. Why?

$$f(\mathbf{x}) = f(x_1, x_2) = -x_1 + 3x_2 + 7.$$

Exercise: Is a linear function affine? Is an affine function linear?



Linear and Affine Functions

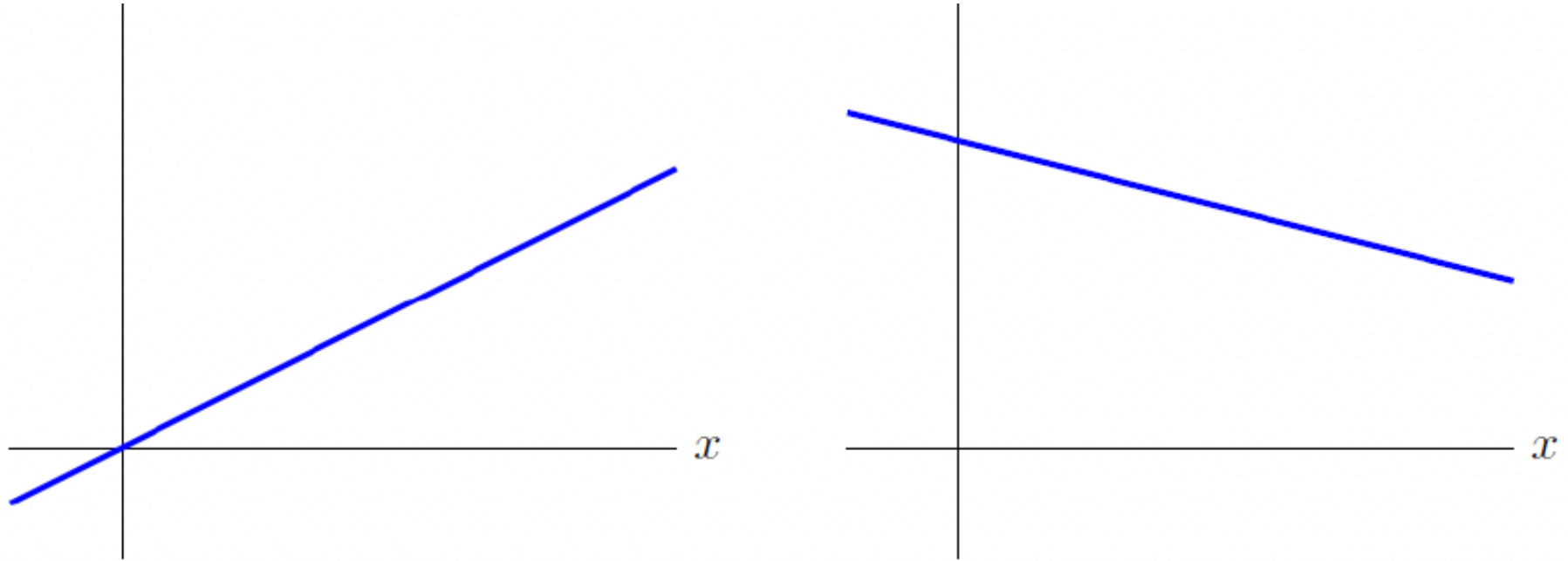


Figure: Left: Linear; Right: Affine but not linear.



Local and global extrema

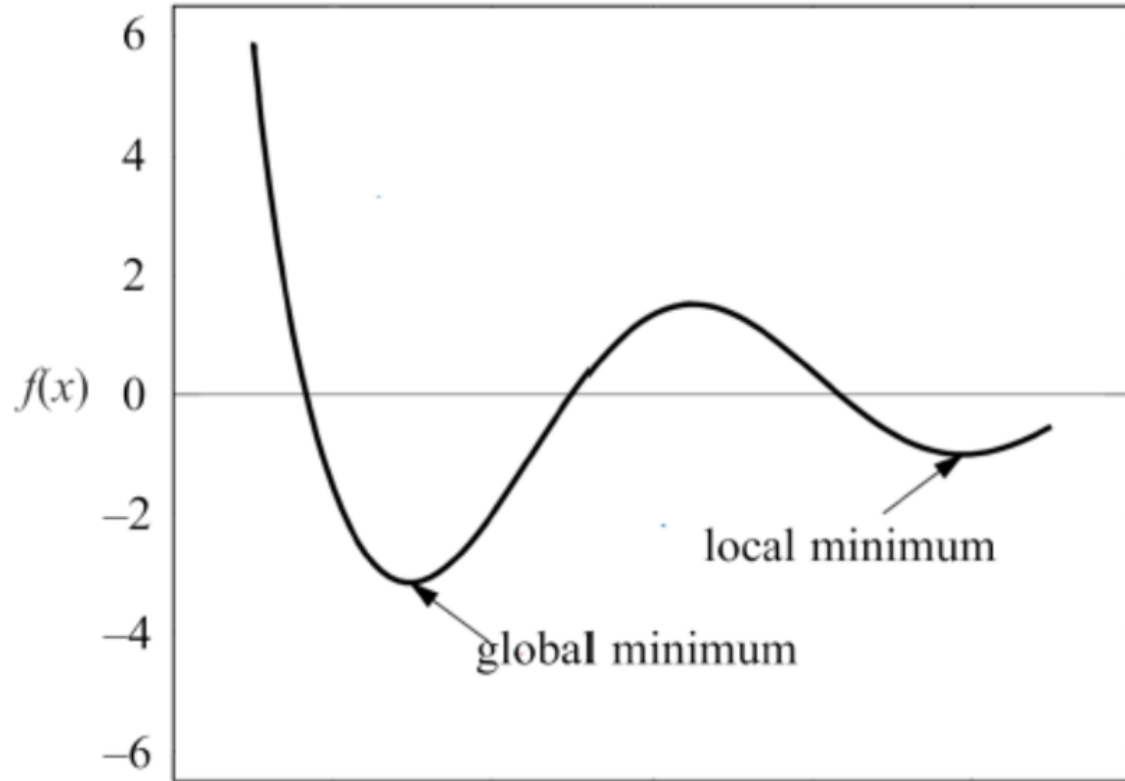
- Consider a function $f : [a, b] \rightarrow \mathbb{R}$.
- The function f has a **local minimum** at $c \in \mathbb{R}$ if $f(x) \geq f(c)$ for all x in an open neighborhood of c .
- The function f has a **global minimum** at $c \in \mathbb{R}$ if $f(x) \geq f(c)$ for all $x \in [a, b]$.

Exercise: If c is a local minimum of f , is it a global minimum? If c is a global minimum of f , is it a local minimum?

Exercise: How would you define **local maximum** and **global maximum**?



Local and global extrema



min and arg min

- For a function $f : X \rightarrow Y$, the **minimum** $\min_{x \in X} f(x)$ returns the smallest value among all elements in the set $\{f(x) : x \in X\}$.
- For a function $f : X \rightarrow Y$, the **argmin** $x^* = \arg \min_{x \in X} f(x)$ returns the value of $x \in X$ that minimizes $f(x)$, i.e.,

$$f(x^*) = \min_{x \in X} f(x)$$

- arg min returns a value from the **domain** of the function X and min returns from the **range** (codomain) Y of the function.
- Let $X = \{0, 1\}$ and $f(0) = \pi$ and $f(1) = e$. Then

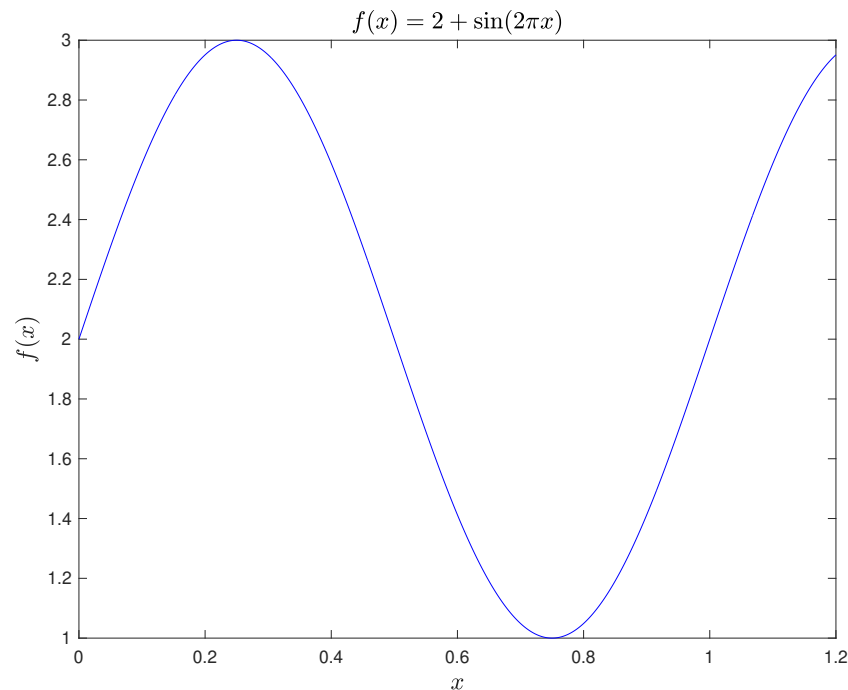
$$\arg \min_{x \in X} f(x) = 1 \qquad \min_{x \in X} f(x) = e,$$

and

$$\arg \max_{x \in X} f(x) = 0 \qquad \max_{x \in X} f(x) = \pi.$$



min and arg min



- Let $f : X = [0, 1.2] \rightarrow \mathbb{R}$ be defined as $f(x) = 2 + \sin(2\pi x)$ (see plot above). Then

$$\arg \min_{x \in X} f(x) = 3/4 \quad \min_{x \in X} f(x) = +1$$



Derivatives

- For a multivariable function $f : \mathbb{R}^d \rightarrow \mathbb{R}$, its gradient vector or derivative at $\mathbf{x} \in \mathbb{R}^d$ is the column vector

$$\underline{\nabla_{\mathbf{x}} f(\mathbf{x}) = \frac{df}{d\mathbf{x}} = \begin{bmatrix} \frac{\partial f}{\partial x_1} & \frac{\partial f}{\partial x_2} & \cdots & \frac{\partial f}{\partial x_d} \end{bmatrix}^{\top}}$$

- Recall that $\frac{\partial f}{\partial x_i}$ is the partial derivative of f with respect to the scalar variable x_i .
- For example, if $f(x_1, x_2) = 2x_1^2 + 5x_1x_2 + 3x_2^3$, then

$$\frac{\partial f}{\partial x_1} = 4x_1 + 5x_2 \quad \text{and} \quad \frac{\partial f}{\partial x_2} = 5x_1 + 9x_2^2.$$



Important derivatives

- There are only two derivatives for functions $f : \mathbb{R}^d \rightarrow \mathbb{R}$ that take vectors to scalars you need to know for now.
- For a fixed vector $\mathbf{a} \in \mathbb{R}^d$, consider $f(\mathbf{x}) = \mathbf{a}^\top \mathbf{x}$ (known as the **inner or dot product**).
Then


$$\nabla_{\mathbf{x}} f(\mathbf{x}) = \mathbf{a}.$$

-
- For a fixed matrix $\mathbf{A} \in \mathbb{R}^{d \times d}$, consider $f(\mathbf{x}) = \mathbf{x}^\top \mathbf{A} \mathbf{x}$ (known as the **quadratic form**).
Then

$$\nabla_{\mathbf{x}} f(\mathbf{x}) = (\mathbf{A} + \mathbf{A}^\top) \mathbf{x}.$$

- In most applications, $\mathbf{A}$ is a **symmetric** matrix (i.e., $\mathbf{A} = \mathbf{A}^\top$) so

$$\nabla_{\mathbf{x}} f(\mathbf{x}) = 2\mathbf{A}\mathbf{x}.$$

- This generalizes the basic fact that if $f(x) = ax^2$, then $\frac{df}{dx} = 2ax$.



Important derivatives

Exercise: Show from the definition of $f(\mathbf{x}) = \mathbf{a}^\top \mathbf{x}$ that

$$\nabla_{\mathbf{x}} f(\mathbf{x}) = \mathbf{a}.$$

Exercise: Show from the definition of $f(\mathbf{x}) = \mathbf{x}^\top \mathbf{A} \mathbf{x}$ that

$$\nabla_{\mathbf{x}} f(\mathbf{x}) = (\mathbf{A} + \mathbf{A}^\top) \mathbf{x}.$$

You'll be forgiven for having to exhibit a substantial amount of meticulous bookkeeping here.

Advice: It is difficult to remember a lot of derivative formulas of complicated multivariate functions. Usually, one consults the **Matrix Cookbook**

<https://www2.imm.dtu.dk/pubdb/edoc/imm3274.pdf>



Outline

- 1 Logistics
- 2 Introduction to Machine Learning
 - Definition and taxonomy
- 3 Mathematical tools**
 - Set and function
 - Probability and estimation**
 - Systems of linear equations



Statistical machine learning deals with uncertainty



Definitions of events, probabilities, joint probabilities and conditional probabilities



Notations

- Random variables (r.v.s): upper (e.g., X)
- Values r.v.s take on: lower case (e.g., x, x_i)
- Sets (in which the r.v.s assume their values): calligraphic font (e.g., $\mathcal{X}$)
- Probability, expectation and variance operators: $\Pr(\cdot)$, $\mathbb{E}[\cdot]$, $\text{Var}(\cdot)$



Discrete random variables

Random variables X and Y :

- X can only take values in the finite set $\mathcal{X} = \{x_1, x_2, \dots, x_M\}$.
- Y can only take values in the finite set $\mathcal{Y} = \{y_1, y_2, \dots, y_L\}$.

A random variable is said to be **discrete** if the set of values it can take has a finite number of values.

$$\sum_{i=1}^M \Pr(X = x_i) = 1, \sum_{i=1}^L \Pr(Y = y_i) = 1.$$



Event

Consider m trials (of sampling X and Y) and let the number of trials for which $X = x_i$ and $Y = y_j$ be m_{ij} . Then, if m is large, we can assume that

$$\Pr(X = x_i, Y = y_j) = \frac{m_{ij}}{m}.$$

- The argument of $\Pr(\cdot)$ is known as an **event**.
- **Event**: A set in which a r.v. (or multiples r.v.s) assume some value(s) in some set.
e.g., $\{X = x_i, Y = y_j\}$.



Sum rule and product rule

- **Sum rule:**

$$\Pr(X = x_i) = \sum_{j=1}^L \Pr(X = x_i, Y = y_j), \quad \Pr(Y = y_j) = \sum_{i=1}^M \Pr(X = x_i, Y = y_j).$$

- **Product rule:**

$$\Pr(X = x_i, Y = y_j) = \Pr(X = x_i | Y = y_j) \cdot \Pr(Y = y_j).$$



Sum rule and product rule

- Sum rule:

$$\Pr(X = x_i) = \sum_{j=1}^L \Pr(X = x_i, Y = y_j), \quad \Pr(Y = y_j) = \sum_{i=1}^M \Pr(X = x_i, Y = y_j).$$

- Product rule:

$$\Pr(X = x_i, Y = y_j) = \Pr(X = x_i | Y = y_j) \cdot \Pr(Y = y_j).$$

$$\Pr(X = x) = \sum_{y \in \mathcal{Y}} \Pr(X = x, Y = y).$$

$$\Pr(X = x, Y = y) = \Pr(X = x | Y = y) \cdot \Pr(Y = y).$$



Bayes' rule

- Probability mass function (pmf):

$$p(x) := \Pr(X = x), \quad p(x, y) := \Pr(X = x, Y = y), \quad p(x|y) := \Pr(X = x|Y = y).$$

- Bayes' rule/Bayes' Theorem:

$$p(y|x) = \frac{p(x, y)}{p(x)} = \frac{p(x|y)p(y)}{p(x)} = \frac{p(x|y)p(y)}{\sum_{y' \in \mathcal{Y}} p(x|y')p(y')}.$$

- ▶ 'Invert the ~~casual~~^{causal} relationship' between X and Y .
- ▶ Fix x , $p(y|x) \propto p(x|y)p(y)$.
- ▶ posterior = $\frac{\text{likelihood} \times \text{prior}}{\text{model evidence}} \propto \text{likelihood} \times \text{prior}.$



Bayes' rule

$$p(y|x) = \frac{p(x|y)p(y)}{\sum_{y' \in \mathcal{Y}} p(x|y')p(y')} \propto p(x|y)p(y) \text{ (when } x \text{ is fixed)}$$

- **Prior:** beliefs or knowledge about y before observing the data x
- **Likelihood:** how likely the observed x is, given a particular value of y
- **Posterior:** updated belief about y after incorporating the prior and the likelihood of the observed data



Continuous random variables

- **Cumulative distribution function (cdf):** $x \in \mathbb{R} \mapsto F_X(x) = \Pr(X \leq x)$ is differentiable on $\mathbb{R}$.
- **Probability density function (pdf):**

↳ 小于等于x的概率, 即从 $-\infty$ 到x的面积

$$f_X(x) = \frac{d}{dx} F_X(x), \quad x \in \mathbb{R}; \quad f_X(x) \geq 0 \quad \forall x \in \mathbb{R} \quad \text{and} \quad \int_{\mathbb{R}} f_X(x) dx = 1.$$



Continuous random variables

- **Cumulative distribution function (cdf):** $x \in \mathbb{R} \mapsto F_X(x) = \Pr(X \leq x)$ is differentiable on $\mathbb{R}$.
- **Probability density function (pdf):**

$$f_X(x) = \frac{d}{dx} F_X(x), \quad x \in \mathbb{R}; \quad f_X(x) \geq 0 \quad \forall x \in \mathbb{R} \text{ and } \int_{\mathbb{R}} f_X(x) dx = 1.$$

- **Joint probability density function** of continuous r.v.s.:

$$f_{X,Y}(x, y) = \frac{\partial^2}{\partial x \partial y} F_{X,Y}(x, y), (x, y) \in \mathbb{R}^2.$$

- Conditional probability and independence can be defined analogously to the discrete case:

$$\Pr(a < X \leq b, c < Y \leq d) = \int_c^d \int_a^b f_{X,Y}(x, y) dx dy.$$



Independence

Definition 3.2 (Independence)

Two random variables X and Y are independent if $p_{X,Y}(x,y) = p_X(x)p_Y(y)$ for all $x \in \mathcal{X}$ and $y \in \mathcal{Y}$.

- X and Y are independent if $p_{X|Y}(x|y) = p_X(x)$ for all $x \in \mathcal{X}$ and $y \in \mathcal{Y}$ such that $p_Y(y) > 0$ (what if $p_Y(y) = 0$?).



Expectation and variance

• Expectation:

▶ Discrete r.v.: $\mathbb{E}X = \sum_{x \in \mathcal{X}} x p_X(x)$.

▶ Continuous r.v.: $\mathbb{E}X = \int_{\mathbb{R}} x f_X(x) dx$.

If g is a function from the domain of X to $\mathbb{R}$, we can obtain the expectation of $Y = g(X)$ in the same way:

$$\mathbb{E}Y = \mathbb{E}[g(X)] = \int_{\mathbb{R}} y f_Y(y) dy = \int_{\mathbb{R}} g(x) f_X(x) dx.$$

▶ **Linearity of expectation:** if $g(X) = aX + b$ (for constants $a, b \in \mathbb{R}$), then $\mathbb{E}[g(X)] = a\mathbb{E}[X] + b = g(\mathbb{E}[X])$. $\Rightarrow E[aX+b] = aE[X] + b$

• Variance:

▶ Discrete r.v.: $\text{Var}(X) = \sum_{x \in \mathcal{X}} (x - \mathbb{E}X)^2 p_X(x)$.

▶ Continuous r.v.: $\text{Var}(X) = \int_{\mathbb{R}} (x - \mathbb{E}X)^2 f_X(x) dx$.

$$\text{Var}[X] = E[X^2] - (E[X])^2$$



Function of random variable

If g is a function from the domain of X to $\mathbb{R}$, we can obtain the expectation of $Y = g(X)$ in the same way.

Proof. I. Discrete case:

$$\mathbb{E}Y = \mathbb{E}[g(X)] = \sum_y yp_Y(y) = \sum_{i=1}^m g(x)p_X(x).$$



Function of random variable

If g is a function from the domain of X to $\mathbb{R}$, we can obtain the expectation of $Y = g(X)$ in the same way.

Proof. I. Discrete case:

$$\mathbb{E}Y = \mathbb{E}[g(X)] = \sum_y yp_Y(y) = \sum_{i=1}^m g(x_i)p_X(x_i).$$

can be shown as below:

$$\sum_{i=1}^m g(x_i)p_X(x_i) = \sum_y \sum_{x:g(x)=y} g(x) \Pr(X = x) = \sum_y \sum_{x:g(x)=y} y \Pr(X = x)$$



Function of random variable

If g is a function from the domain of X to $\mathbb{R}$, we can obtain the expectation of $Y = g(X)$ in the same way.

Proof. I. Discrete case:

$$\mathbb{E}Y = \mathbb{E}[g(X)] = \sum_y y p_Y(y) = \sum_{i=1}^m g(x_i) p_X(x_i).$$

can be shown as below:

$$\begin{aligned} \sum_{i=1}^m g(x_i) p_X(x_i) &= \sum_y \sum_{x: g(x)=y} g(x) \Pr(X = x) = \sum_y \sum_{x: g(x)=y} y \Pr(X = x) \\ &= \sum_y \sum_{x: g(x)=y} y \Pr(Y = y, X = x) = \sum_y y \sum_{x: g(x)=y} \Pr(Y = y, X = x) \end{aligned}$$



Function of random variable

If g is a function from the domain of X to $\mathbb{R}$, we can obtain the expectation of $Y = g(X)$ in the same way.

Proof. I. Discrete case:

$$\mathbb{E}Y = \mathbb{E}[g(X)] = \sum_y yp_Y(y) = \sum_{i=1}^m g(x_i)p_X(x_i).$$

can be shown as below:

$$\begin{aligned}\sum_{i=1}^m g(x_i)p_X(x_i) &= \sum_y \sum_{x:g(x)=y} g(x) \Pr(X = x) = \sum_y \sum_{x:g(x)=y} y \Pr(X = x) \\ &= \sum_y \sum_{x:g(x)=y} y \Pr(Y = y, X = x) = \sum_y y \sum_{x:g(x)=y} \Pr(Y = y, X = x) \\ &= \sum_y y \Pr(Y = y) = \sum_y yp_Y(y) = \mathbb{E}Y = \mathbb{E}[g(X)].\end{aligned}$$



Function of random variable

Proof (continued). II. Continuous case:

$$\mathbb{E}Y = \mathbb{E}[g(X)] = \int_{\mathbb{R}} yf_Y(y)dy = \int_{\mathbb{R}} g(x)f_X(x)dx.$$

Refer to Theorem 4.1.1 in the book ‘ Probability and Statistics’.



Function of random variable

If g is a function from the domain of X to $\mathbb{R}$, we can obtain the expectation of $Y = g(X)$ in the same way.

$$\text{(Discrete case)} \quad \mathbb{E}Y = \mathbb{E}[g(X)] = \sum_y yp_Y(y) = \sum_{i=1}^m g(x)p_X(x),$$

$$\text{(Continuous case)} \quad \mathbb{E}Y = \mathbb{E}[g(X)] = \int_{\mathbb{R}} yf_Y(y)dy = \int_{\mathbb{R}} g(x)f_X(x)dx.$$



Maximum likelihood estimation (MLE)

选一组参数使得“已经观察到的数据”出现的概率最大

- In machine learning, we almost always do **NOT** have access to the underlying distributions that the data are generated from.
- Rather, we have access to sample data and we would like to **use it to estimate some parameters**.



Maximum likelihood estimation: Gaussian distribution

- If samples in $\mathcal{D} = \{X_1, \dots, X_m\}$ are independently drawn from the **univariate Gaussian distribution**

$$f_X(x; \mu, \sigma^2) = \mathcal{N}(x; \mu, \sigma^2) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{1}{2\sigma^2}(x - \mu)^2\right) \quad x \in \mathbb{R}, \quad (3.1)$$

we can use the samples in $\mathcal{D}$ to estimate the parameter vector $\theta = (\mu, \sigma^2)$.

- Note: notation $f_X(x; \mu, \sigma^2)$ or $f_X(x; \theta)$ emphasize that the density is parametrized by the parameters (μ, σ^2) or θ .



Maximum likelihood estimation: Gaussian distribution

- If samples in $\mathcal{D} = \{X_1, \dots, X_m\}$ are independently drawn from the **univariate Gaussian distribution**

$$f_X(x; \mu, \sigma^2) = \mathcal{N}(x; \mu, \sigma^2) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{1}{2\sigma^2}(x - \mu)^2\right) \quad x \in \mathbb{R}, \quad (3.1)$$

we can use the samples in $\mathcal{D}$ to estimate the parameter vector $\theta = (\mu, \sigma^2)$.

- Note: notation $f_X(x; \mu, \sigma^2)$ or $f_X(x; \theta)$ emphasize that the density is parametrized by the parameters (μ, σ^2) or θ .

- ⊙ Maximum likelihood estimates for the mean and variance:

$$\hat{\mu}_{\text{ML}} = \frac{1}{m} \sum_{i=1}^m X_i \quad \text{and} \quad \hat{\sigma}_{\text{ML}}^2 = \frac{1}{m} \sum_{i=1}^m (X_i - \hat{\mu})^2. \quad (3.2)$$



$$L(\mu, \sigma^2) = \prod_{i=1}^n \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{1}{2\sigma^2}(x_i - \mu)^2\right)$$

取对数 *

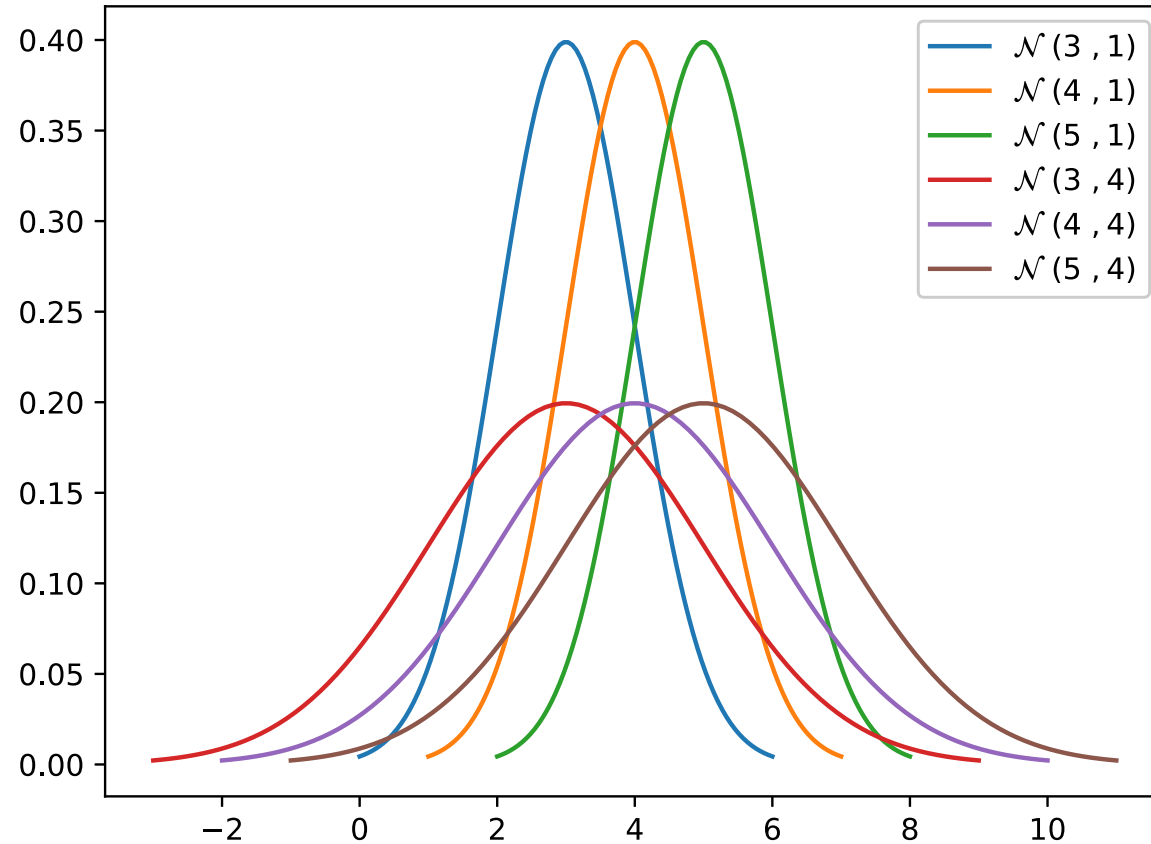
$$\ln L(\mu, \sigma^2) = -\frac{n}{2} \ln(2\pi) - \frac{n}{2} \ln(\sigma^2) - \frac{1}{2\sigma^2} \sum_{i=1}^n (x_i - \mu)^2$$

求偏导

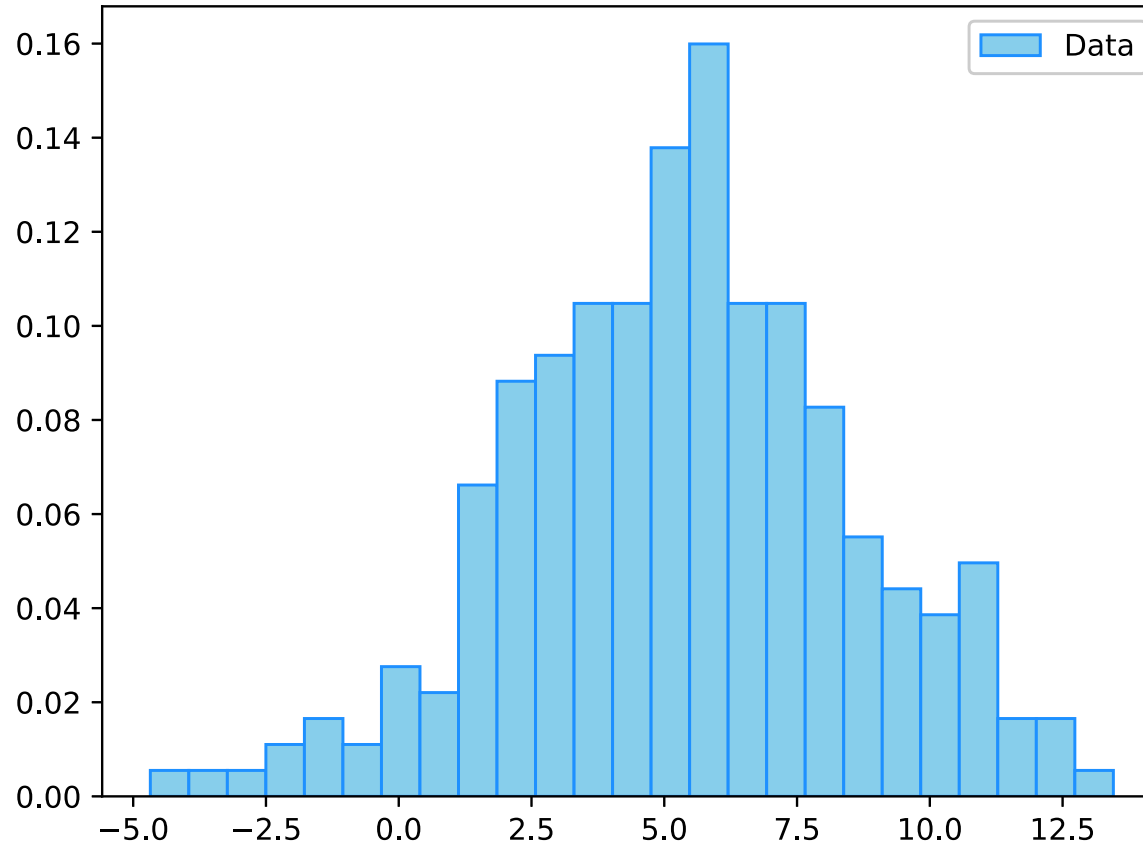
$$\frac{\partial \ln L}{\partial \mu} = \frac{1}{\sigma^2} \sum_{i=1}^n (x_i - \mu) = 0 \Rightarrow \underline{\hat{\mu}} = \frac{1}{n} \sum_{i=1}^n x_i$$

$$\frac{\partial \ln L}{\partial \sigma^2} = -\frac{n}{2\sigma^2} + \frac{1}{2(\sigma^2)^2} \sum_{i=1}^n (x_i - \mu)^2 = \underline{\hat{\sigma}^2} = \frac{1}{n} \sum_{i=1}^n (x_i - \hat{\mu})^2$$

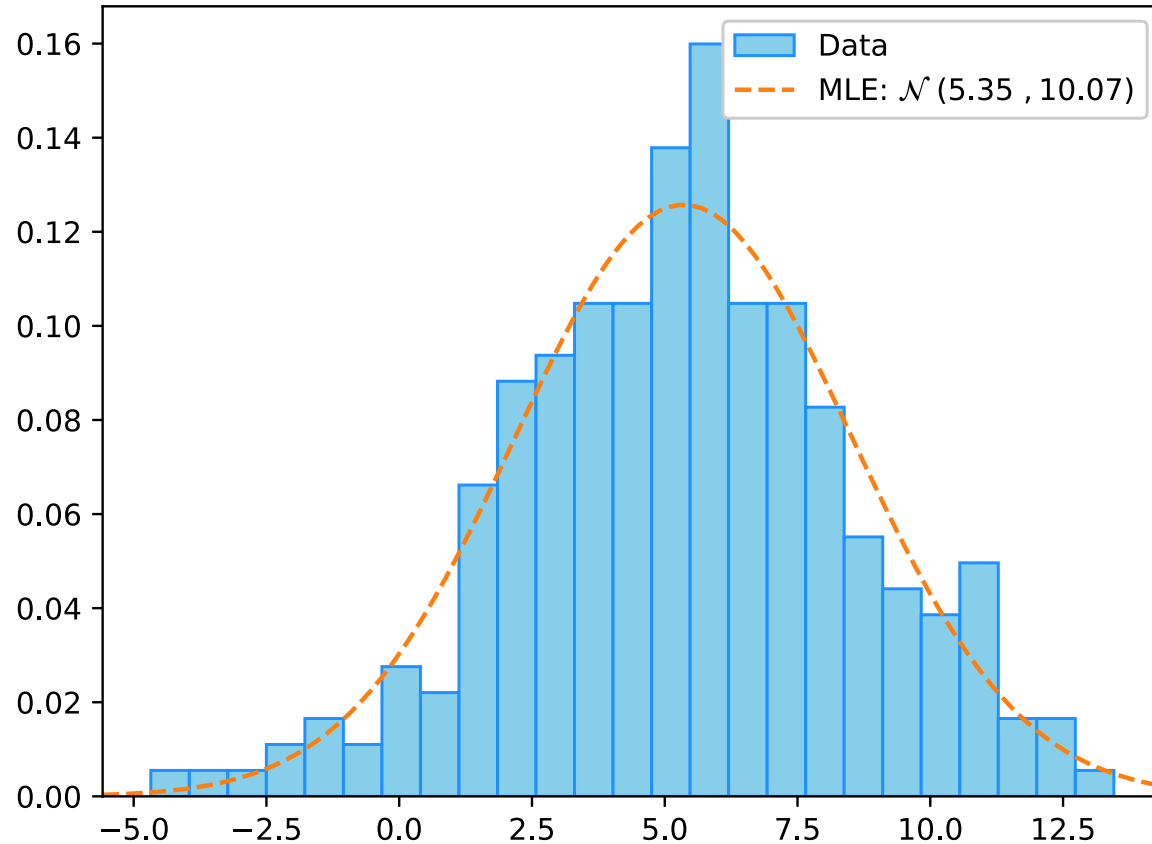
Maximum likelihood estimation: Gaussian distribution



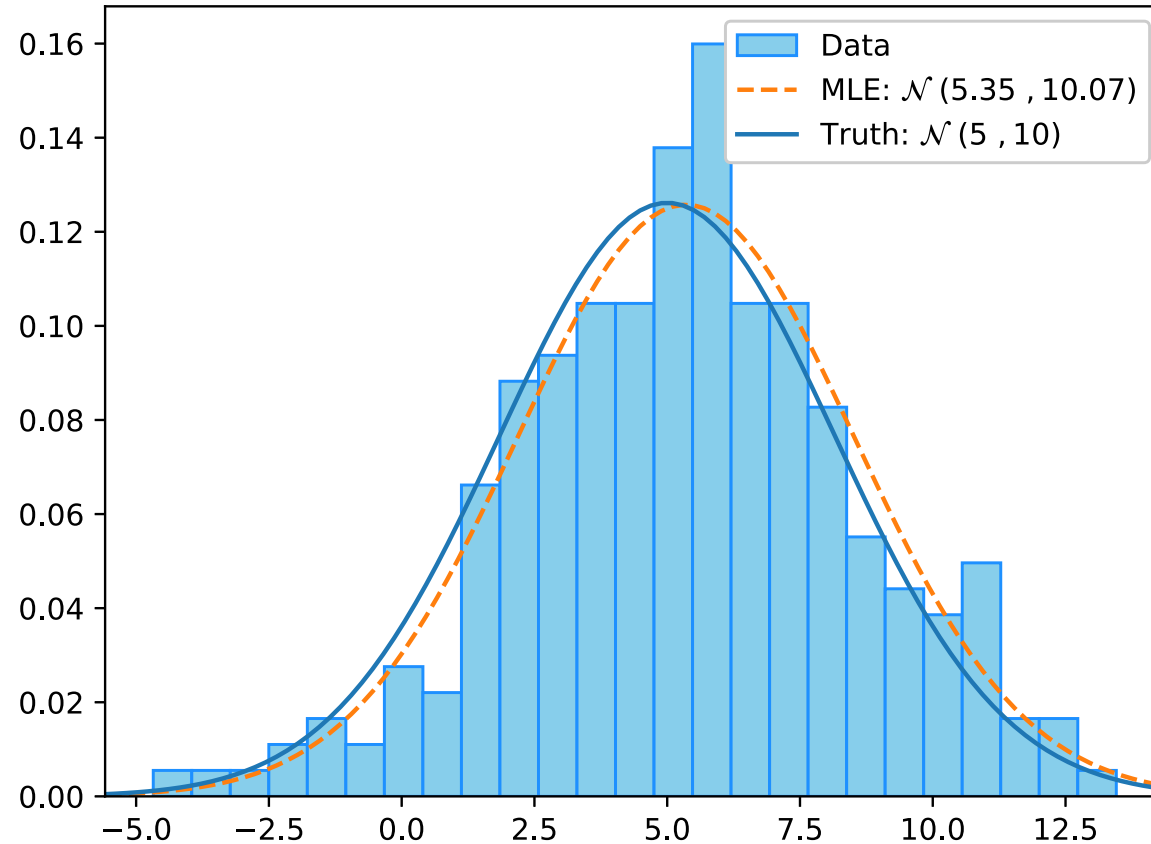
Maximum likelihood estimation: Gaussian distribution



Maximum likelihood estimation: Gaussian distribution



Maximum likelihood estimation: Gaussian distribution



Maximum likelihood estimation: Bernoulli distribution

例 1

- Samples in $\mathcal{D} = \{X_1, \dots, X_m\}$ are generated independently from the Bernoulli (coin toss) distribution

$$p_X(x; \theta) = \begin{cases} 1 - \theta & x = 0 \\ \theta & x = 1 \end{cases} \quad (3.3)$$

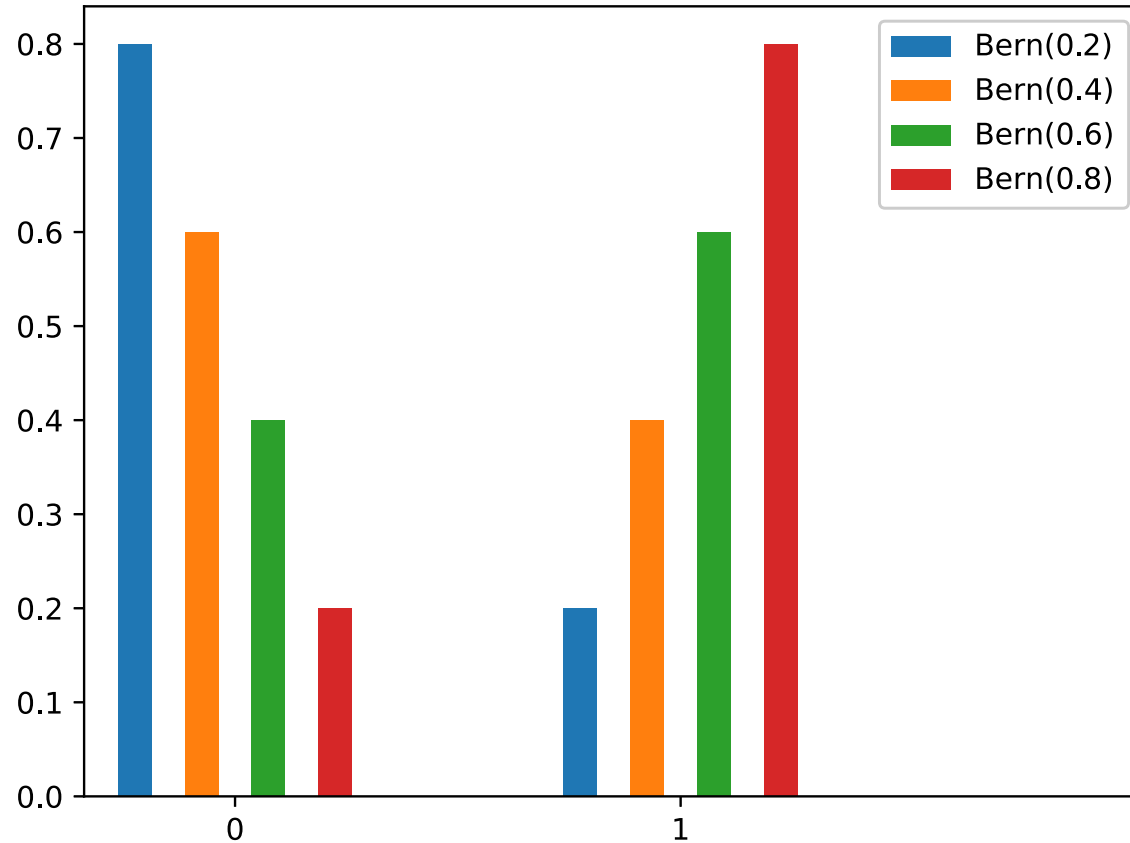
or

$$p_X(x; \theta) = (1 - \theta)^{1-x} \theta^x, \quad x \in \{0, 1\}. \quad (3.4)$$

- How would we estimate θ from samples?



Maximum likelihood estimation: Bernoulli distribution



Maximum likelihood estimation: Bernoulli distribution

- $M_1 := \sum_{i=1}^m X_i$ denote the total number of ones in $\mathcal{D}$. Consider

$$\hat{\theta}_{\text{ML}} = \arg \max_{\theta \in (0,1)} \prod_{i=1}^m p_X(X_i; \theta) \quad (3.5)$$



Maximum likelihood estimation: Bernoulli distribution

- $M_1 := \sum_{i=1}^m X_i$ denote the total number of ones in $\mathcal{D}$. Consider

$$\hat{\theta}_{\text{ML}} = \arg \max_{\theta \in (0,1)} \prod_{i=1}^m p_X(X_i; \theta) \quad (3.5)$$

$$= \arg \max_{\theta \in (0,1)} \sum_{i=1}^m \log p_X(X_i; \theta) \quad (3.6)$$



Maximum likelihood estimation: Bernoulli distribution

- $M_1 := \sum_{i=1}^m X_i$ denote the total number of ones in $\mathcal{D}$. Consider

$$\hat{\theta}_{\text{ML}} = \arg \max_{\theta \in (0,1)} \prod_{i=1}^m p_X(X_i; \theta) \quad (3.5)$$

$$= \arg \max_{\theta \in (0,1)} \sum_{i=1}^m \log p_X(X_i; \theta) \quad (3.6)$$

$$= \arg \max_{\theta \in (0,1)} \sum_{i=1}^m [(1 - X_i) \log(1 - \theta) + X_i \log \theta] \quad (3.7)$$

$$= \arg \max_{\theta \in (0,1)} (m - M_1) \log(1 - \theta) + M_1 \log \theta. \quad (3.8)$$



Maximum likelihood estimation: Bernoulli distribution

- $M_1 := \sum_{i=1}^m X_i$ denote the total number of ones in $\mathcal{D}$. Consider

$$\hat{\theta}_{\text{ML}} = \arg \max_{\theta \in (0,1)} \underbrace{(m - M_1) \log(1 - \theta) + M_1 \log \theta}_{\text{Objective function } g(\theta)}. \quad (3.9)$$



Maximum likelihood estimation: Bernoulli distribution

- $M_1 := \sum_{i=1}^m X_i$ denote the total number of ones in $\mathcal{D}$. Consider

$$\hat{\theta}_{\text{ML}} = \arg \max_{\theta \in (0,1)} \underbrace{(m - M_1) \log(1 - \theta) + M_1 \log \theta}_{\text{Objective function } g(\theta)}. \quad (3.9)$$

- $g(\theta)$ is strictly concave $\Rightarrow$ differentiating and setting to zero yields the (unique) maximum:



Maximum likelihood estimation: Bernoulli distribution

- $M_1 := \sum_{i=1}^m X_i$ denote the total number of ones in $\mathcal{D}$. Consider

$$\hat{\theta}_{\text{ML}} = \arg \max_{\theta \in (0,1)} \underbrace{(m - M_1) \log(1 - \theta) + M_1 \log \theta}_{\text{Objective function } g(\theta)}. \quad (3.9)$$

- $g(\theta)$ is strictly concave $\Rightarrow$ differentiating and setting to zero yields the (unique) maximum:

$$g'(\theta) = -\frac{m - M_1}{1 - \theta} + \frac{M_1}{\theta}, \quad (3.10)$$

$$\hat{\theta}_{\text{ML}} = \arg \max_{\theta \in (0,1)} g(\theta) \implies g'(\hat{\theta}_{\text{ML}}) = 0 \implies \hat{\theta}_{\text{ML}} = \frac{M_1}{m}. \quad (3.11)$$



Maximum likelihood estimation: Bernoulli distribution

- $M_1 := \sum_{i=1}^m X_i$ denote the total number of ones in $\mathcal{D}$. Consider

$$\hat{\theta}_{\text{ML}} = \arg \max_{\theta \in (0,1)} \underbrace{(m - M_1) \log(1 - \theta) + M_1 \log \theta}_{\text{Objective function } g(\theta)}. \quad (3.9)$$

- $g(\theta)$ is strictly concave $\Rightarrow$ differentiating and setting to zero yields the (unique) maximum:

$$g'(\theta) = -\frac{m - M_1}{1 - \theta} + \frac{M_1}{\theta}, \quad (3.10)$$

$$\hat{\theta}_{\text{ML}} = \arg \max_{\theta \in (0,1)} g(\theta) \implies g'(\hat{\theta}_{\text{ML}}) = 0 \implies \hat{\theta}_{\text{ML}} = \frac{M_1}{m}. \quad (3.11)$$

- ◉ Mean θ is estimated by the empirical mean M_1/m : agrees with common sense!



Maximum likelihood estimation: exponential distribution

例 2

- Samples in $\mathcal{D} = \{X_1, \dots, X_m\}$ are generated independently from the exponential distribution (memory-less)

$$f_X(x; \theta) = \begin{cases} \theta \exp(-\theta x) & x \geq 0 \\ 0 & x < 0 \end{cases}.$$

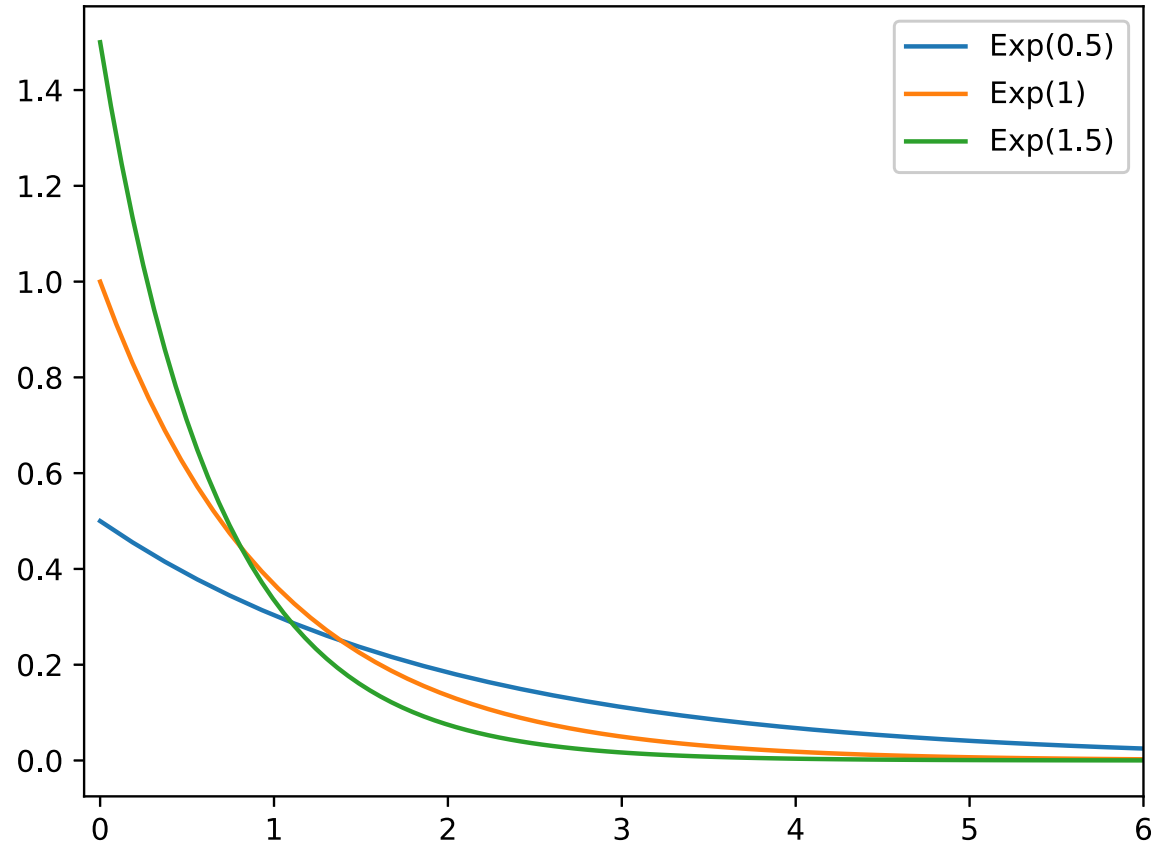
- $\theta > 0$ is the unknown (rate) parameter. In fact, $\mathbb{E}[X] = 1/\theta$.
- Case: network request arrival time

$$F_X(x; \theta) = \begin{cases} 1 - \exp(-\theta x) & x \geq 0 \\ 0 & x < 0 \end{cases}.$$

- How would we estimate θ from samples?



Maximum likelihood estimation: exponential distribution



Maximum likelihood estimation: exponential distribution

- Consider

$$\hat{\theta}_{\text{ML}} = \arg \max_{\theta > 0} \prod_{i=1}^m f_X(X_i; \theta) = \arg \max_{\theta > 0} \sum_{i=1}^m \log f_X(X_i; \theta) \quad (3.12)$$



Maximum likelihood estimation: exponential distribution

- Consider

$$\hat{\theta}_{\text{ML}} = \arg \max_{\theta > 0} \prod_{i=1}^m f_X(X_i; \theta) = \arg \max_{\theta > 0} \sum_{i=1}^m \log f_X(X_i; \theta) \quad (3.12)$$

$$= \arg \max_{\theta > 0} \sum_{i=1}^m (\log \theta - \theta X_i) = \arg \max_{\theta > 0} m \log \theta - \sum_{i=1}^m \theta X_i. \quad (3.13)$$



Maximum likelihood estimation: exponential distribution

- Consider

$$\hat{\theta}_{\text{ML}} = \arg \max_{\theta > 0} \prod_{i=1}^m f_X(X_i; \theta) = \arg \max_{\theta > 0} \sum_{i=1}^m \log f_X(X_i; \theta) \quad (3.12)$$

$$= \arg \max_{\theta > 0} \sum_{i=1}^m (\log \theta - \theta X_i) = \arg \max_{\theta > 0} m \log \theta - \sum_{i=1}^m \theta X_i. \quad (3.13)$$

- $g(\theta) = m \log \theta - \sum_{i=1}^m \theta X_i$ is strictly concave $\Rightarrow$ differentiating and setting to zero yields the (unique) maximum:



Maximum likelihood estimation: exponential distribution

- Consider

$$\hat{\theta}_{\text{ML}} = \arg \max_{\theta > 0} \prod_{i=1}^m f_X(X_i; \theta) = \arg \max_{\theta > 0} \sum_{i=1}^m \log f_X(X_i; \theta) \quad (3.12)$$

$$= \arg \max_{\theta > 0} \sum_{i=1}^m (\log \theta - \theta X_i) = \arg \max_{\theta > 0} m \log \theta - \sum_{i=1}^m \theta X_i. \quad (3.13)$$

- $g(\theta) = m \log \theta - \sum_{i=1}^m \theta X_i$ is strictly concave $\Rightarrow$ differentiating and setting to zero yields the (unique) maximum:

$$g'(\theta) = \frac{m}{\theta} - \sum_{i=1}^m X_i, \quad (3.14)$$

$$g'(\hat{\theta}_{\text{ML}}) = 0 \implies \hat{\theta}_{\text{ML}} = \hat{\theta}_{\text{ML}} = \frac{m}{\sum_{i=1}^m X_i} = \left(\frac{1}{m} \sum_{i=1}^m X_i \right)^{-1}. \quad (3.15)$$



Maximum likelihood estimation: exponential distribution

- Exponential distribution (memoryless distribution)

$$f_X(x; \theta) = \begin{cases} \theta \exp(-\theta x) & x \geq 0 \\ 0 & x < 0 \end{cases}.$$

- $\mathbb{E}[X] = 1/\theta$, $\hat{\theta}_{\text{ML}} = \left(\frac{1}{m} \sum_{i=1}^m X_i\right)^{-1}$.



Maximum likelihood estimation: exponential distribution

- Exponential distribution (memoryless distribution)

$$f_X(x; \theta) = \begin{cases} \theta \exp(-\theta x) & x \geq 0 \\ 0 & x < 0 \end{cases}.$$

- $\mathbb{E}[X] = 1/\theta$, $\hat{\theta}_{\text{ML}} = \left(\frac{1}{m} \sum_{i=1}^m X_i\right)^{-1}$.
- $\odot$ $\frac{1}{m} \sum_{i=1}^m X_i$ is the empirical mean:
it seems plausible that $m / \left(\sum_{i=1}^m X_i\right)$ is a “good” estimate of θ .
- Why would it not be “good”?



Maximum likelihood estimation

$$\hat{\theta}_{\text{ML}} = \arg \max_{\theta \in (0,1)} p_{X_1, X_2, \dots, X_m} (X_1, X_2, \dots, X_m; \theta)$$



Maximum likelihood estimation

$$\hat{\theta}_{\text{ML}} = \arg \max_{\theta \in (0,1)} \underbrace{p_{X_1, X_2, \dots, X_m}(X_1, X_2, \dots, X_m; \theta)}_{\text{Likelihood}}$$



Maximum likelihood estimation

$$\begin{aligned}\hat{\theta}_{\text{ML}} &= \arg \max_{\theta \in (0,1)} \underbrace{p_{X_1, X_2, \dots, X_m}(X_1, X_2, \dots, X_m; \theta)}_{\text{Likelihood}} \\ &= \arg \max_{\theta \in (0,1)} \prod_{i=1}^m p_X(X_i; \theta) \quad (\text{when } X_i\text{'s are independent})\end{aligned}$$



Maximum likelihood estimation

$$\begin{aligned}\hat{\theta}_{\text{ML}} &= \arg \max_{\theta \in (0,1)} \underbrace{p_{X_1, X_2, \dots, X_m}(X_1, X_2, \dots, X_m; \theta)}_{\text{Likelihood}} \\ &= \arg \max_{\theta \in (0,1)} \prod_{i=1}^m p_X(X_i; \theta) \quad (\text{when } X_i\text{'s are independent})\end{aligned}$$

- **Likelihood**: probability of observing the dataset $\mathcal{D}$ given a set of parameters θ



Maximum likelihood estimation

$$\begin{aligned}\hat{\theta}_{\text{ML}} &= \arg \max_{\theta \in (0,1)} \underbrace{p_{X_1, X_2, \dots, X_m}(X_1, X_2, \dots, X_m; \theta)}_{\text{Likelihood}} \\ &= \arg \max_{\theta \in (0,1)} \prod_{i=1}^m p_X(X_i; \theta) \quad (\text{when } X_i\text{'s are independent})\end{aligned}$$

- **Likelihood**: probability of observing the dataset $\mathcal{D}$ given a set of parameters θ
- **Maximum**: seeks the set of parameters θ that maximizes this likelihood function, i.e., makes the observed dataset as “likely” as possible under the model



Maximum likelihood estimation

$$\begin{aligned}\hat{\theta}_{\text{ML}} &= \arg \max_{\theta \in (0,1)} \underbrace{p_{X_1, X_2, \dots, X_m} (X_1, X_2, \dots, X_m; \theta)}_{\text{Likelihood}} \\ &= \arg \max_{\theta \in (0,1)} \prod_{i=1}^m p_X (X_i; \theta) \quad (\text{when } X_i\text{'s are independent})\end{aligned}$$

- Likelihood: probability of observing the dataset $\mathcal{D}$ given a set of parameters θ
- Maximum: seeks the set of parameters θ that maximizes this likelihood function, i.e., makes the observed dataset as “likely” as possible under the model
- Concept of MLE: linear regression, logistic regression, ...



Maximum likelihood estimation (MLE)

- **Consistency.** as the sample size increases to infinity, the estimator will converge to the true parameter value: $\hat{\theta}_{\text{ML}} \xrightarrow{p} \theta$ as $n \rightarrow \infty$.
 - ▶ We say $X_n \xrightarrow{p} X$ as $n \rightarrow \infty$ if

$$\lim_{n \rightarrow \infty} \Pr(|X_n - X| > \varepsilon) = 0 \quad \forall \varepsilon > 0.$$



Maximum likelihood estimation (MLE)

- **Consistency.** as the sample size increases to infinity, the estimator will converge to the true parameter value: $\hat{\theta}_{\text{ML}} \xrightarrow{p} \theta$ as $n \rightarrow \infty$.

- ▶ We say $X_n \xrightarrow{p} X$ as $n \rightarrow \infty$ if

$$\lim_{n \rightarrow \infty} \Pr(|X_n - X| > \varepsilon) = 0 \quad \forall \varepsilon > 0.$$

- **Unbiasedness.** MLE is not necessarily unbiased, but in certain cases, it can be unbiased.
 - ▶ We say an estimator of a given parameter is unbiased if its expected value is equal to the true value of the parameter.



Maximum likelihood estimation (MLE)

- **Efficiency.** MLE is asymptotically efficient in large samples, i.e., it achieves the lowest possible variance among all unbiased estimators.
 - ▶ Cramér-Rao Lower Bound (CRLB): theoretical lower bound for the variance of any unbiased estimator.



Maximum likelihood estimation (MLE)

- **Efficiency.** MLE is asymptotically efficient in large samples, i.e., it achieves the lowest possible variance among all unbiased estimators.
 - ▶ Cramér-Rao Lower Bound (CRLB): theoretical lower bound for the variance of any unbiased estimator.
- **Asymptotic Normality.** MLE is asymptotically normal:
$$\sqrt{n}(\hat{\theta}_{\text{ML}} - \theta) \xrightarrow{d} \mathcal{N}(0, I(\theta)^{-1}).$$
 - ▶ $I(\theta)$: Fisher information of θ (larger information $\implies$ smaller variance).
 - ▶ We say $X_n \xrightarrow{d} X$ as $n \rightarrow \infty$ if

$$\lim_{n \rightarrow \infty} F_{X_n}(x) = F_X(x) \quad \forall x \text{ at which } F_X(x) \text{ is continuous.}$$



Maximum likelihood estimation (MLE)

- **Efficiency.** MLE is asymptotically efficient in large samples, i.e., it achieves the lowest possible variance among all unbiased estimators.
 - ▶ Cramér-Rao Lower Bound (CRLB): theoretical lower bound for the variance of any unbiased estimator.

- **Asymptotic Normality.** MLE is asymptotically normal:

$$\sqrt{n}(\hat{\theta}_{\text{ML}} - \theta) \xrightarrow{d} \mathcal{N}(0, I(\theta)^{-1}).$$

- ▶ $I(\theta)$: Fisher information of θ (larger information $\implies$ smaller variance).
- ▶ We say $X_n \xrightarrow{d} X$ as $n \rightarrow \infty$ if

$$\lim_{n \rightarrow \infty} F_{X_n}(x) = F_X(x) \quad \forall x \text{ at which } F_X(x) \text{ is continuous.}$$

- **Model Sensitivity.** MLE is sensitive to model assumptions, and incorrect assumptions can lead to biased or inconsistent estimates.



Maximum likelihood estimation (MLE)

- **Consistency.** as the sample size increases to infinity, the estimator will converge to the true parameter value: $\hat{\theta}_{\text{ML}} \xrightarrow{P} \theta$ as $n \rightarrow \infty$. 数据多了趋近真实值
- **Unbiasedness.** MLE is not necessarily unbiased, but in certain cases, it can be unbiased.
- **Efficiency.** MLE is asymptotically efficient in large samples, i.e., it achieves the lowest possible variance among all unbiased estimators. 大样本下方差小
- **Asymptotic Normality.** MLE is asymptotically normal:
$$\sqrt{n}(\hat{\theta}_{\text{ML}} - \theta) \xrightarrow{d} \mathcal{N}(0, I(\theta)^{-1}).$$
- **Model Sensitivity.** MLE is sensitive to model assumptions, and incorrect assumptions can lead to biased or inconsistent estimates.



Outline

- 1 Logistics
- 2 Introduction to Machine Learning
 - Definition and taxonomy
- 3 Mathematical tools**
 - Set and function
 - Probability and estimation
 - Systems of linear equations**



Systems of linear equations

Definition 3.3 (Rules)

A vector space over the reals consists of a set $\mathcal{V}$, a vector sum operation $+: \mathcal{V} \times \mathcal{V} \rightarrow \mathcal{V}$ and a scalar multiplication operation $\cdot: \mathbb{R} \times \mathcal{V} \rightarrow \mathcal{V}$ satisfying the following properties.

- Commutativity: $\mathbf{x} + \mathbf{y} = \mathbf{y} + \mathbf{x}$ for all $\mathbf{x}, \mathbf{y} \in \mathcal{V}$;
- Associativity: $(\mathbf{x} + \mathbf{y}) + \mathbf{z} = \mathbf{x} + (\mathbf{y} + \mathbf{z})$ for all $\mathbf{x}, \mathbf{y}, \mathbf{z} \in \mathcal{V}$;
- Identity element of addition: $\mathbf{x} + \mathbf{0} = \mathbf{x}$ for all $\mathbf{x} \in \mathcal{V}$;
- Inverse element of addition: For every $\mathbf{x} \in \mathcal{V}$, there exists an element $-\mathbf{x} \in \mathcal{V}$ such that $\mathbf{x} + (-\mathbf{x}) = \mathbf{0}$;
- Associativity of scalar multiplication: For all $a, b \in \mathbb{R}$ and $\mathbf{x} \in \mathcal{V}$, $a(b\mathbf{x}) = (ab)\mathbf{x}$;
- Identity element of scalar multiplication: $1\mathbf{x} = \mathbf{x}$ for $\mathbf{x} \in \mathcal{V}$;
- Distributivity of scalar multiplication with respect to vector addition: $a(\mathbf{x} + \mathbf{y}) = a\mathbf{x} + a\mathbf{y}$ for all $a \in \mathbb{R}$ and $\mathbf{x}, \mathbf{y} \in \mathcal{V}$;
- Distributivity of scalar multiplication with respect to addition in $\mathbb{R}$: $(a + b)\mathbf{x} = a\mathbf{x} + b\mathbf{x}$ for all $a, b \in \mathbb{R}$ and $\mathbf{x} \in \mathcal{V}$.



Definition 3.4 (Linear independence)

A set of vectors $\{\mathbf{x}_1, \mathbf{x}_2, \dots, \mathbf{x}_k\}$ from a vector space $\mathcal{V}$ is linearly independent if

$$\beta_1 \mathbf{x}_1 + \beta_2 \mathbf{x}_2 + \dots + \beta_k \mathbf{x}_k = \mathbf{0} \quad (3.16)$$

implies that $\beta_1 = \beta_2 = \dots = \beta_k = 0$.

- $\{\mathbf{x}_1, \mathbf{x}_2, \dots, \mathbf{x}_k\}$ are linearly independent $\iff$ no vector $\mathbf{x}_i$ can be expressed as a linear combination of the other vectors $\{\mathbf{x}_1, \dots, \mathbf{x}_{i-1}, \mathbf{x}_{i+1}, \dots, \mathbf{x}_k\}$.



Matrix

Definition 3.5 (Basis of vector space)

A set of vectors $\{\mathbf{x}_1, \mathbf{x}_2, \dots, \mathbf{x}_k\}$ is a basis for a vector space $\mathcal{V}$ if

- $\mathcal{V} = \text{span}\{\mathbf{x}_1, \mathbf{x}_2, \dots, \mathbf{x}_k\}$;
- $\{\mathbf{x}_1, \mathbf{x}_2, \dots, \mathbf{x}_k\}$ is a **linearly independent set** of vectors.

Equivalently, every $\mathbf{x} \in \mathcal{V}$ can be uniquely written as $\sum_{i=1}^k \beta_i \mathbf{x}_i$ for some $\{\beta_i\}_{i=1}^k \subset \mathbb{R}$. The number of vectors in any basis of $\mathcal{V}$ is called the **dimension** of $\mathcal{V}$, written as $\dim(\mathcal{V})$.

Definition 3.6 (nullspace and range of matrix)

The **nullspace (also called kernel)** of a matrix $\mathbf{A} \in \mathbb{R}^{m \times d}$ is defined as

$$\mathcal{N}(\mathbf{A}) = \{\mathbf{x} \in \mathbb{R}^d : \mathbf{A}\mathbf{x} = \mathbf{0}\}. \quad (3.17)$$

The range or column space of $\mathbf{A}$ is defined as

$$\mathcal{R}(\mathbf{A}) = \{\mathbf{A}\mathbf{x} : \mathbf{x} \in \mathbb{R}^f\} \subset \mathbb{R}^m. \quad (3.18)$$



Matrix

Definition 3.7 (Rank of matrix)

The **column rank or rank** of a matrix $\mathbf{A} \in \mathbf{R}^{m \times d}$ is defined as

$$\text{rank}(\mathbf{A}) = \dim(\mathcal{R}(\mathbf{A})). \quad (3.19)$$

In other words, the column rank of $\mathbf{A}$ is the dimension of the column space of $\mathbf{A}$.



Matrix

Definition 3.7 (Rank of matrix)

The **column rank or rank** of a matrix $\mathbf{A} \in \mathbf{R}^{m \times d}$ is defined as

$$\text{rank}(\mathbf{A}) = \dim(\mathcal{R}(\mathbf{A})). \quad (3.19)$$

In other words, the column rank of $\mathbf{A}$ is the dimension of the column space of $\mathbf{A}$.

- ⊙ **Rank-nullity:** $\text{rank}(\mathbf{A}) + \dim(\mathcal{N}(\mathbf{A})) = d$.
- It is always true that $\text{rank}(\mathbf{A}) \leq \min\{m, d\}$.
- A matrix is **full rank** if $\text{rank}(\mathbf{A}) = \min\{m, d\}$.
- A matrix is **full column rank (resp. full row rank)** if the set of columns (resp. rows) of the matrix is linearly independent.
- If the matrix $\mathbf{A}$ is square (i.e., $m = d$) and it is full rank, then the inverse $\mathbf{A}^{-1}$ exists.



Solutions to linear systems

Solve systems of equations of the form ($\underline{x}_i$, $1 \leq i \leq d$: columns of $\mathbf{X}$)

$$\mathbf{X}\mathbf{w} = \mathbf{y} \quad \text{or} \quad [\underline{x}_1 \ \underline{x}_2 \ \dots \ \underline{x}_d] \begin{bmatrix} w_1 \\ w_2 \\ \vdots \\ w_d \end{bmatrix} = \begin{bmatrix} y_1 \\ y_2 \\ \vdots \\ y_d \end{bmatrix}, \quad \text{where } \underline{x}_i = \begin{bmatrix} x_{1,i} \\ x_{2,i} \\ \vdots \\ x_{m,i} \end{bmatrix}. \quad (3.20)$$

Matrix $\mathbf{X} \in \mathbb{R}^{m \times d}$ and vector $\mathbf{y} \in \mathbb{R}^m$ are given and $\mathbf{w} \in \mathbb{R}^d$ is to be found.



Solutions to linear systems

Solve systems of equations of the form ($\underline{x}_i$, $1 \leq i \leq d$: columns of $\mathbf{X}$)

$$\mathbf{X}\mathbf{w} = \mathbf{y} \quad \text{or} \quad [\underline{x}_1 \ \underline{x}_2 \ \dots \ \underline{x}_d] \begin{bmatrix} w_1 \\ w_2 \\ \vdots \\ w_d \end{bmatrix} = \begin{bmatrix} y_1 \\ y_2 \\ \vdots \\ y_d \end{bmatrix}, \quad \text{where } \underline{x}_i = \begin{bmatrix} x_{1,i} \\ x_{2,i} \\ \vdots \\ x_{m,i} \end{bmatrix}. \quad (3.20)$$

Matrix $\mathbf{X} \in \mathbb{R}^{m \times d}$ and vector $\mathbf{y} \in \mathbb{R}^m$ are given and $\mathbf{w} \in \mathbb{R}^d$ is to be found.

- $m = d$: if matrix $\mathbf{X}$ is square and full rank, $\mathbf{X}^{-1}$ exists $\implies$ solve for $\mathbf{w}$ by simple matrix inversion and multiplication $\mathbf{w} = \mathbf{X}^{-1}\mathbf{y}$.



Solutions to linear systems

Solve systems of equations of the form ($\underline{x}_i$, $1 \leq i \leq d$: columns of $\mathbf{X}$)

$$\mathbf{X}\mathbf{w} = \mathbf{y} \quad \text{or} \quad [\underline{x}_1 \ \underline{x}_2 \ \dots \ \underline{x}_d] \begin{bmatrix} w_1 \\ w_2 \\ \vdots \\ w_d \end{bmatrix} = \begin{bmatrix} y_1 \\ y_2 \\ \vdots \\ y_d \end{bmatrix}, \quad \text{where } \underline{x}_i = \begin{bmatrix} x_{1,i} \\ x_{2,i} \\ \vdots \\ x_{m,i} \end{bmatrix}. \quad (3.20)$$

Matrix $\mathbf{X} \in \mathbb{R}^{m \times d}$ and vector $\mathbf{y} \in \mathbb{R}^m$ are given and $\mathbf{w} \in \mathbb{R}^d$ is to be found.

- $m = d$: if matrix $\mathbf{X}$ is square and full rank, $\mathbf{X}^{-1}$ exists $\implies$ solve for $\mathbf{w}$ by simple matrix inversion and multiplication $\mathbf{w} = \mathbf{X}^{-1}\mathbf{y}$.
- $m \neq d$: **existence and uniqueness of solutions to the linear system?**

Augmented matrix $\tilde{\mathbf{X}} = [\mathbf{X} \ \mathbf{y}] \in \mathbb{R}^{m \times (d+1)} : \text{rank}(\mathbf{X}) \leq \text{rank}(\tilde{\mathbf{X}})$.



Solutions to linear systems

- ◎ **Method:** apply **Guassian-elimination** method to the augmented matrix $\tilde{\mathbf{X}}$.

Theorem 3.8 (Rouché-Capelli Theorem)

- *The system in (3.20) admits a unique solution if and only if $\text{rank}(\mathbf{X}) = \text{rank}(\tilde{\mathbf{X}}) = d$;*
- *The system in (3.20) has no solution if and only if $\text{rank}(\mathbf{X}) < \text{rank}(\tilde{\mathbf{X}})$;*
- *The system in (3.20) has infinitely many solutions if and only if $\text{rank}(\mathbf{X}) = \text{rank}(\tilde{\mathbf{X}}) < d$.*

